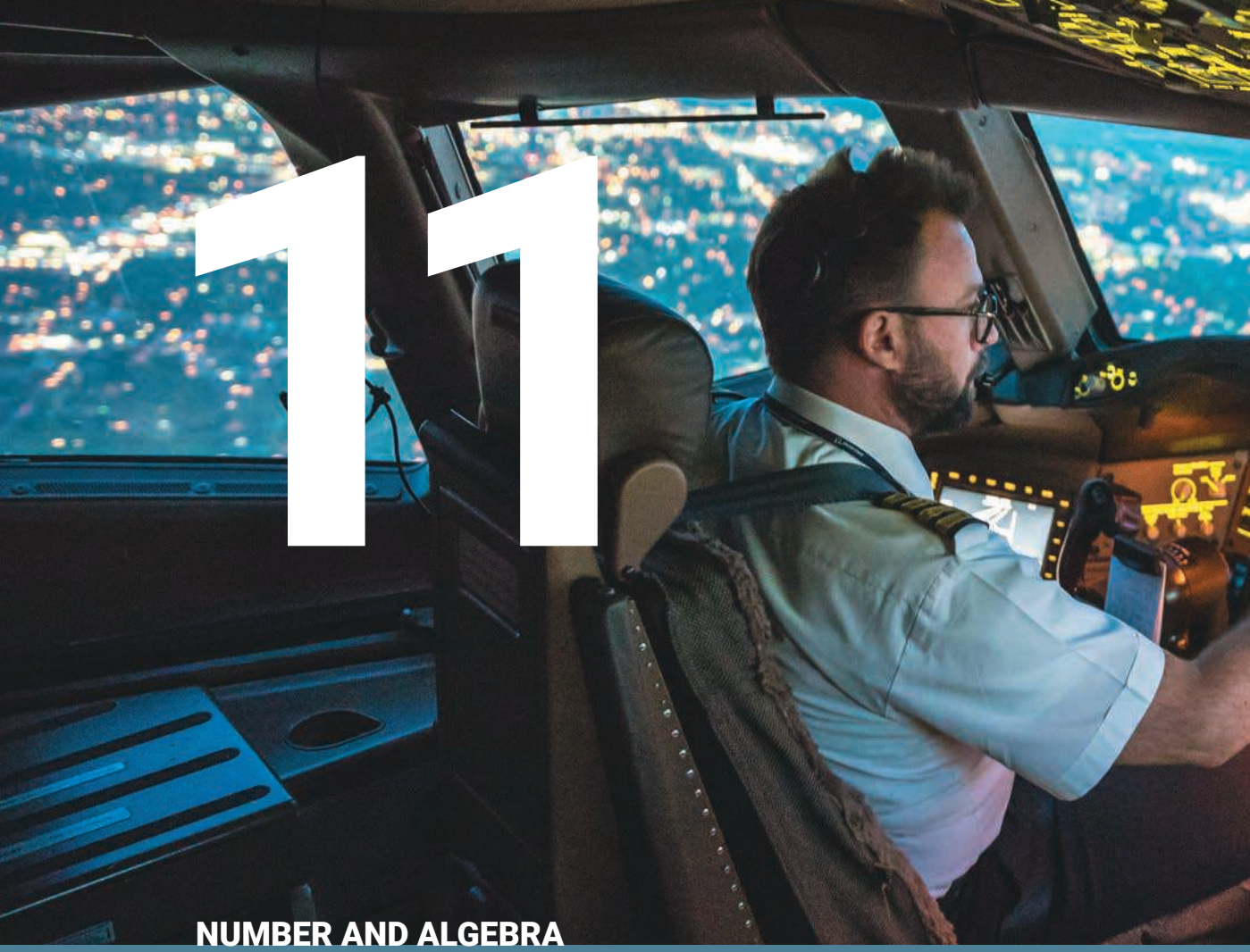

11

NUMBER AND ALGEBRA

GRAPHING LINEAR EQUATIONS

The famous French philosopher and mathematician René Descartes (1596–1650) is famous for discovering the connections between algebra and geometry. His concept of coordinate geometry connected algebra and geometry using the graphs of straight lines and curves. In computer modelling, the use of coordinates allows us to locate things more precisely, especially in three-dimensional (3D) space. Number plane graphs are used to track changes over short and long periods of time, such as the height and speed of an aeroplane.

Chapter outline

		Working mathematically				
11.01	Tables of values	U	F			
11.02	Finding the rule	U	F		R	C
11.03	Finding rules for number patterns	U	F	PS	R	C
11.04	The number plane	U	F		R	C
11.05	Graphing number patterns	U	F		R	C
11.06	Graphing linear equations	U	F		R	C
11.07	Finding the equation of a line	U	F		R	C
11.08	Comparing linear equations	U	F		R	C
11.09	The gradient of a line*	U	F		R	C
11.10	The linear equation $y = mx + c$*	U	F		R	C
11.11	Applying linear equations	U	F	PS	R	C
11.12	Solving linear equations graphically	U	F			
11.13	Intersecting lines	U	F		R	

*YEAR 9 EXTENSION

Wordbank

consecutive numbers Numbers that follow each other in order, for example, 2, 3, 4

constant term The term in an equation that is a number only and does not contain a variable, for example, the 5 in $y = 2x + 5$

horizontal Going across, sideways, flat

linear Involving a line

linear equation A formula whose graph is a straight line

number plane A coordinate grid system based on the x- and y-axes, also called a Cartesian plane

table of values A table of ordered pairs of numbers, usually following a formula and which can be graphed on a number plane

vertical Going up and down, at a right angle to the horizontal

y-intercept The y-value at which a line crosses the y-axis on the number plane

Quiz
Wordbank 11

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (10):

11.02 Finding the rule
11.04 The number plane
11.05 Graphing number patterns
11.06 Linear functions • Graphing linear equations
11.07 Finding the equation of a line • Horizontal and vertical lines
11.08 Linear functions
11.09 The gradient of a line • Gradient (slope) of a line
11.10 Graphing $y = mx + c$

Twig video (1):

11.04 Coordinate geometry: Descartes

PhET interactives (4):

11.02 Function builder: Basics • Function builder
11.06 Graphing lines
11.08, 11.10 Graphing slope-intercept

Skillsheet (1):

11.04 The number plane

Quizzes (5):

- Wordbank 11
- SkillCheck 11
- Mental skills 11
- Language of maths 11
- Test yourself 11

Worksheets (17):

11.01 Tables of values
11.03 Patterns and rules
11.04 The number plane • Plane grid 2 • Number plane review • Number plane grid paper • A page of number planes • Plotting points and lines • Number plane writing activity
11.05 Number plane grid paper • A page of number planes • Graphing tables of values
11.06 Number plane grid paper • A page of lines • A page of number planes
11.07 Finding linear equations
11.09 Drawing gradients • A page of lines • Gradient between two points
11.09, 11.10, 11.12, 11.13 Number plane grid paper • A page of number planes
11.11 Number plane grid paper • 1 cm grid paper
11.13 Line intersections
Mind map Graphing linear equations

Puzzles (6):

11.04 Coordinates code puzzle
11.06 Matching linear equations • Graphing functions
11.07, 11.08 Matching linear equations
11.09 Find the gradient 1
11.10 Equation of a line
Language of maths Linear equations crossword

Technology (2):

11.04 Plotting points
11.06 Graphing straight lines: Finding intercepts

Presentation (1):

11.04 Graph coordinates

Spreadsheets (2):

11.04 Points plotter
11.07 Linear functions plotter

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- complete a table of values to graph a linear equation on a number plane
- identify and describe a number pattern, then represent it algebraically and graphically on a number plane
- determine the formula for a table of values and set of points on a number plane
- compare and contrast the graphs of different linear equations
- (EXTENSION, YEAR 9 CORE) find the gradient of an interval or line
- (EXTENSION, YEAR 9 CORE) use the gradient-intercept equation of a straight line $y = mx + c$

- solve problems that can be modelled by linear equations
- solve linear equations graphically
- graph 2 intersecting lines and identify the point of intersection

SkillCheck ANSWERS ON P. 605

Quiz
SkillCheck 11

11.01

1 For each equation, find the missing number.

a $2 \times 3 + ___ = 10$ **b** $2 \times (-1) + ___ = 3$ **c** $3 \times 1 - ___ = 1$

d $-2 \times 2 + ___ = -2$ **e** $4 \times 3 - ___ = 8$ **f** $-1 \times (-1) - ___ = -4$

2 Evaluate each expression.

a $3x - 2$ if $x = 2$ **b** $\frac{1}{2}x + 3$ if $x = 6$ **c** $2x - 1$ if $x = -1$

d $-x + 1$ if $x = 0$ **e** $-2x - 1$ if $x = 1$ **f** $10 - x$ if $x = 3$

3 Solve each equation.

a $3x - 2 = 7$ **b** $-2x - 1 = 3$ **c** $-x + 1 = 4$

d $2x - 1 = -5$ **e** $3x + 2 = x - 1$ **f** $x - 2 = 4 - x$

Tables of values

11.01

In order to graph straight lines, we need points. In Year 7, you were given a **table of values** to plot on the number plane, or used a rule or formula to create a table of values.

Worksheet
Tables of values

Example 1

Complete this table of values using the formula $v = 2u + 1$.

u	−2	−1	0	1	2	6	9
v							

Solution

Substitute each value of u in the table into the formula $v = 2u + 1$ to find the value of v.

- When $u = -2$, $v = 2 \times (-2) + 1 = -3$
- When $u = -1$, $v = 2 \times (-1) + 1 = -1$
- When $u = 0$, $v = 2 \times 0 + 1 = 1$
- When $u = 1$, $v = 2 \times 1 + 1 = 3$
- When $u = 2$, $v = 2 \times 2 + 1 = 5$
- When $u = 6$, $v = 2 \times 6 + 1 = 13$
- When $u = 9$, $v = 2 \times 9 + 1 = 19$

Completing the table, we have:

u	−2	−1	0	1	2	6	9
v	−3	−1	1	3	5	13	19

EXERCISE 11.01 ANSWERS ON P. 605

Tables of values U F

EXAMPLE 1

1 Complete each table of values given the formula.

a $y = x - 1$

x	3	7	−1	−8	4	5
y					3	

b $q = 3p$

p	8	−3	2	5	10	−4
q				15		

c $k = h \div 2$

h	12	8	−2	−6	0	7
k	6					

d $y = x + 7$

x	−2	11	7	0	13	−9
y		18				

e $b = 2a$

a	8	−1	−5	0	4	2
b			−10			

f $y = x - 3$

x	10	−4	7	5	−8	11
y	7					

2 Given the rule $y = -x - 4$, what is the value of y when $x = -3$? Select **A**, **B**, **C** or **D**.

A -1 **B** -7 **C** 7 **D** 12

3 Complete each table of values given the formula.

a $p = 5m - 1$

m	3	10	1	4	8	−6
p		49				

b $q = 3p - 2$

p	4	−1	7	10	2	−6
q			19			

c $t = 4r + 6$

r	2	0	−3	9	−5	4
t	14					

d $z = 5y + 2$

y	4	8	7	−5	6	−2
z		42				

e $y = 2x - 3$

x	0	1	2	3	4	5
y						

f $b = \frac{1}{2}a - 2$

a	−2	−1	0	1	2	3
b						

g $k = 3h + 5$

h	−1	0	1	2	3	4
k						

h $y = 1 - x$

x	−2	−1	0	1	2	3
y						

Foundation Standard Complex

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4 Which table of values matches the formula $y = 3x - 1$? Select **A**, **B**, **C** or **D**.

A

x	1	2	3	4	5
y	2	3	4	5	6

B

x	1	2	3	4	5
y	2	5	8	11	14

C

x	1	2	3	4	5
y	4	7	10	13	16

D

x	1	2	3	4	5
y	0	3	6	9	12

5 Which table of values follows the rule $c = \frac{h+2}{2}$? Select **A**, **B**, **C** or **D**.

A

h	0	2	4
c	1	3	5

B

h	0	2	4
c	1	2	5

C

h	0	2	4
c	0	1	3

D

h	0	2	4
c	1	2	3

Finding the rule

11.02

We can often find the formula for a table of values. If the values in the top row are **consecutive** (increase by 1 each time), look for a pattern in the values in the bottom row.

Example 2

Find the formula for each table of values.

a

p	2	3	4	6	7	8
q	1	2	3	5	6	7

b

m	−2	−1	0	1	2	3
n	−6	−3	0	3	6	9

c

r	1	2	3	4	5	6
t	1	3	5	7	9	11

d

d	1	2	3	4	5	6
e	8	12	16	20	24	28

Video
Finding the rule

Interactives
Function builder: Basics

Function builder

Foundation | Standard | Complex

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Solution

a What has been done to each value in the top row to get the value in the bottom row?

p	2	3	4	6	7	8
q	1	2	3	5	6	7

The pattern is:

$2 - 1 = 1$
$3 - 1 = 2$
$4 - 1 = 3$, and so on.
The formula is $q = p - 1$.

b The pattern is:

m	−2	−1	0	1	2	3
n	−6	−3	0	3	6	9

$3 \times (-2) = -6$
$3 \times (-1) = -3$
$3 \times 0 = 0$, and so on.
The formula is $n = 3m$.

c The formula for this table of values is not so obvious. It involves **2** operations: a multiplication and either an addition or a subtraction.

r	1	2	3	4	5	6
t	1	3	5	7	9	11
		2	2	2	2	2

If the values in the top row are consecutive, the bottom row helps us find the **multiplier**.

The values in the bottom row go up by 2 each time, so the multiplier is 2. This means the formula must have $2 \times r$ in it.

$2 \times 1 - 1 = 1$
$2 \times 2 - 1 = 3$
$2 \times 3 - 1 = 5$, and so on.
The formula is $t = 2r - 1$.

d The values in the top row are **consecutive**, the values in the bottom row go up by 4, so the multiplier is 4, and the equation is of the form $e = 4d$ ____.

d	1	2	3	4	5	6
e	8	12	16	20	24	28
		4	4	4	4	4

Choose any ordered pair from the table to find the missing number, say (2, 12).

When $d = 2$, $e = 4 \times 2 + 4 = 12$, so the missing number is 4.

So the equation is $e = 4d + 4$.

(Checking that this is also true for another ordered pair (1, 8).
When $d = 1$, $e = 4 \times 1 + 4 = 8$)

EXERCISE 11.02 ANSWERS ON P. 605

Finding the rule UFRC

EXAMPLE 2

11.02

1 Copy and complete the formula for each table of values. R C

a $y = ___ x$

x	0	1	2	3
y	0	3	6	9

b $y = ___ x$

x	0	1	2	3
y	0	−2	−4	−6

c $y = x + _____$

x	−1	0	1	2
y	3	4	5	6

d $y = x - _____$

x	0	1	2	3
y	−4	−3	−2	−1

e $y = 10 - _____ x$

x	−1	0	1	2
y	11	10	9	8

f $y = x \div _____$

x	0	1	2	3
y	0	$\frac{1}{2}$	1	$1\frac{1}{2}$

g $k = _____$

p	1	2	3	4
k	5	6	7	8

h $y = _____$

x	−1	0	1	2
y	−4	−3	−2	−1

i $j = _____$

h	4	5	9	11
j	8	10	18	22

j $p = _____$

m	0	5	30	45
p	0	1	6	9

2 Find the formula for each table, then complete the last 2 columns. R C

a

f	1	2	3	4	7	8
h	1	4	7	10		

b

m	1	2	3	4	6	9
p	2	7	12	17		

c

m	0	1	2	3	5	8
b	3	6	9	12		

d

h	3	4	5	6	9	11
k	8	10	12	14		

e

r	0	1	2	3	4	7
s	1	4	7	10		

f

a	2	3	4	5	10	11
b	2	4	6	8		

g

m	0	1	2	3	5	6
n	5	8	11	14		

h

c	3	4	5	6	9	10
d	31	41	51	61		

i

w	3	4	5	6	7	8
x	14	19	24	29		

j

y	5	6	7	8	9	10
z	4	6	8	10		

Foundation Standard Complex

k

a	1	2	3	4	7	9
m	5	9	13	17		

l

z	4	5	6	7	9	12
t	1	3	5	7		

3 What is the formula for this table? Select the correct answer **A**, **B**, **C** or **D**.

p	3	4	5	6
m	8	11	14	17

A $m = 2p + 1$ **B** $m = 3p - 1$ **C** $m = 2p + 4$ **D** $m = 3p + 1$

4 What is the formula for this table? Select **A**, **B**, **C** or **D**.

x	1	2	3	4	5
y	−5	−7	−9	−11	−13

A $y = -2x - 5$ **B** $y = 2x - 5$ **C** $y = -2x - 3$ **D** $y = 2x - 3$

5 Find the formula for each table, then complete the table. R C

a

x	0	1	2	3	9
y	−1	−3	−5	−7	

b

A	0	1	2	3	10
C	0	0.5	1	1.5	

c

s	−1	0	1	2	7
t	18	13	8	3	

d

t	−1	0	1	2	5
v	30	20	10		

e

x	0	1	2	3	6
D	−0.5	−1	−1.5		

11.03 Finding rules for number patterns

Worksheet
Patterns and rules

Example 3

For this number pattern: 2, 5, 8, 11, 14, ...

a find the 8th term

b find a formula for the nth term, T.

Solution

a This number pattern is increasing by 3 each time. By continuing the pattern:

2, 5, 8, 11, 14, 17, 20, 23

8th term = 20 + 3 = 23

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b Write the number pattern as a table of values.

Term of pattern, n	1	2	3	4	5
Number, T	2	5	8	11	14

The number pattern goes up by 3 each time, so the multiplier for the formula is 3, and the equation is of the form $T = 3n$ ____.

Choose any ordered pair from the table to find the missing number, say (1, 2).

When $n = 1$, $T = 3 \times 1 - 1 = 2$, so the missing number is -1.

$3 \times 1 - 1 = 2$

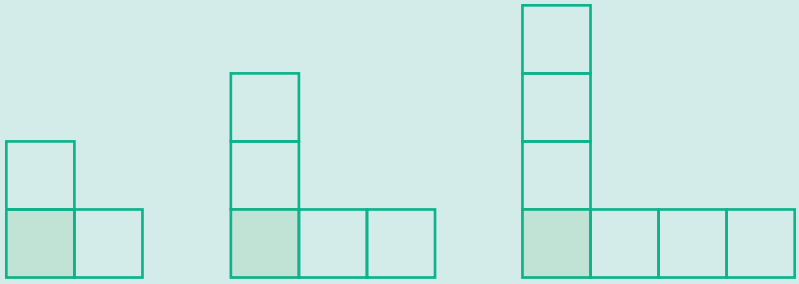
$3 \times 2 - 1 = 5$

$3 \times 3 - 1 = 8$, and so on

So the general formula for the nth term is $T = 3n - 1$.

Example 4

Examine this geometrical pattern.



arm length = 1 arm length = 2 arm length = 3

a Copy and complete this table for the pattern.

Arm length, a	1	2	3	4	5	6
Number of tiles, t						

b Write the rule for this pattern in words.

c Write the rule as a formula.

d How many tiles would be needed to make a design with an arm length of:

i 18? **ii** 60?

Solution

a

Arm length, a	1	2	3	4	5	6
Number of tiles, t	3	5	7	9	11	13

b The number of tiles increases by 2 each time. This means that the multiplier is 2, and the equation is of the form $t = 2a$ ____.

Choose (1, 3) from the table to find the missing number.

When $a = 1$, $t = 2 \times 1 + 1 = 3$.

In words, the rule is: The number of tiles is 2 times the arm length plus 1.

c As a formula, the rule is $t = 2a + 1$.

d **i** When $a = 18$, $t = 2 \times 18 + 1 = 37$

37 tiles are needed for a design with an arm length of 18.

ii When $a = 60$, $t = 2 \times 60 + 1 = 121$

121 tiles are needed for a design with an arm length of 60.

EXERCISE 11.03 ANSWERS ON P. 605

Finding rules for number patterns U F PS R C

EXAMPLE 3

1 Find the formula for the nth term, T, of the number pattern $-1, -3, -5, -7, \ldots$
Select the correct answer **A**, **B**, **C** or **D**. R C

A $T = n - 2$ **B** $T = 1 - 2n$ **C** $T = 2n - 1$ **D** $T = n - 3$

2 For each number pattern: R C

i find the 8th term

ii find a formula for the nth term, T

iii use the formula to find the 8th term and the 20th term.

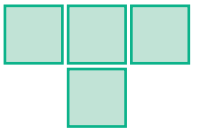
a 1, 3, 5, 7, 9, ... **b** 2, 4, 6, 8, 10, ... **c** 8, 10, 12, 14, 16, ...

d 7, 10, 13, 16, 19, ... **e** 7, 15, 23, 31, 39, ... **f** 2, 7, 12, 17, 22, ...

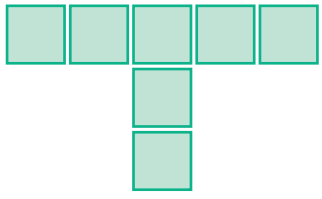
g 19, 17, 15, 13, 11, ... **h** 1, 4, 9, 16, 25, ...

EXAMPLE 4

3 **a** Here are the first 2 T-shapes in a geometrical pattern. Draw the next 3 T-shapes. R C



arm length = 1



arm length = 2

b Copy and complete the table.

Arm length, a	1	2	3	4	5	8	11
Number of tiles, t	4						

c Write the rule for the pattern in words.

d Write the rule as a formula.

e How many tiles are needed to build a T-shape with arm length of:

i 15? **ii** 33?

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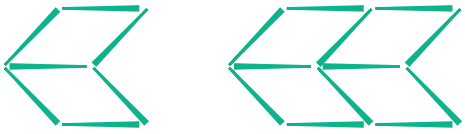
4 **a** Here are the first 2 flight shapes, made from toothpicks. Draw the next 3. R C



flight shape 1 flight shape 2

b Copy and complete the table.

Number of flight shape, f	1	2	3	4	6	9	10
Number of toothpicks, t	7						

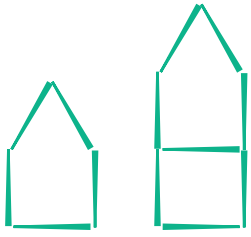
c Write the rule for the pattern in words.

d Write the rule as a formula.

e How many toothpicks are needed to build:

i flight shape 25? **ii** flight shape 40?

5 **a** The first 2 rockets are shown below. Draw the next 3 rockets. R C



stage 1 stage 2

b Copy and complete the table.

Number of stage, s	1	2	3	4	5	8	11
Number of toothpicks, t							

c Write the rule for the pattern in words.

d Write the rule as a formula.

e How many toothpicks are needed to build:

i stage 12? **ii** stage 45?

6 For each matchstick pattern on the next page: R C

i draw the next 3 terms

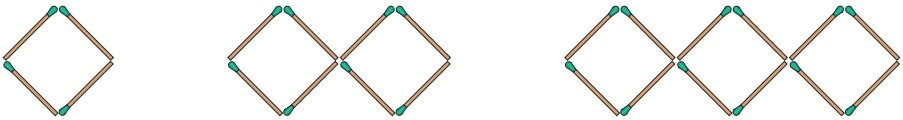
ii copy and complete this table

Number of shapes (n)	1	2	3	4	5
Number of matchsticks (T)					

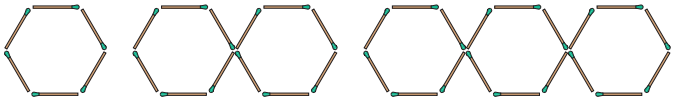
iii find the formula for the pattern

iv find the number of matchsticks required to make 12 shapes.

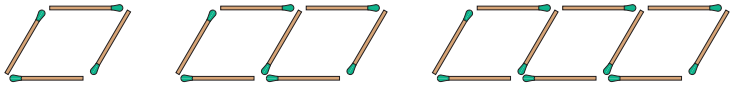
a



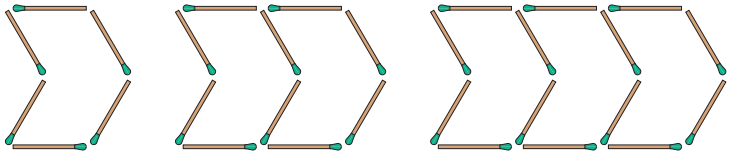
b



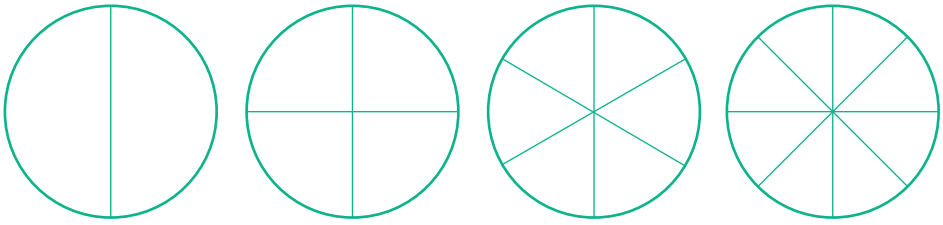
c



d



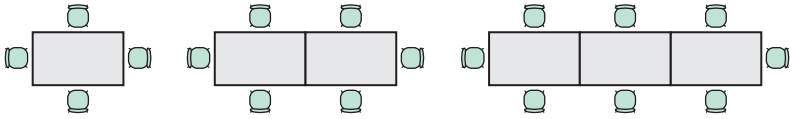
7 The pattern below is formed by cutting a cake c times to make p pieces. Which of the following is the formula that best describes the pattern? Select **A**, **B**, **C** or **D**. R C



1 cut 2 cuts 3 cuts 4 cuts

A $p = 2c$ **B** $c = 2p$ **C** $p = c + 2$ **D** $c = p + 2$

8 In a restaurant, chairs are placed around rectangular tables as shown below. PS R C



a Find the number of chairs required for:

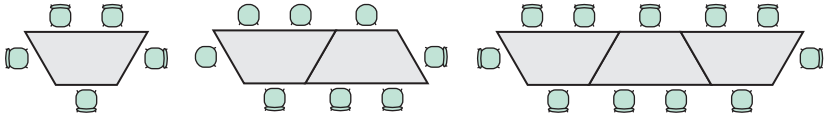
i 2 tables **ii** 3 tables **iii** 4 tables **iv** 12 tables

b Find a formula that relates the number of chairs, C, to the number of tables, T.

c How many chairs would be needed for a row of 50 tables?

d A booking is made at the restaurant for 30 people. How many tables are needed?

9 Students arrange themselves around desks ready for group work in the classroom. The seating arrangement for n students at d desks is shown. PS R C



a Find the number of students who can be seated around a row of:

i 2 desks **ii** 3 desks **iii** 4 desks **iv** 10 desks

b Find a formula that relates the number of students, n, to the number of desks, d.

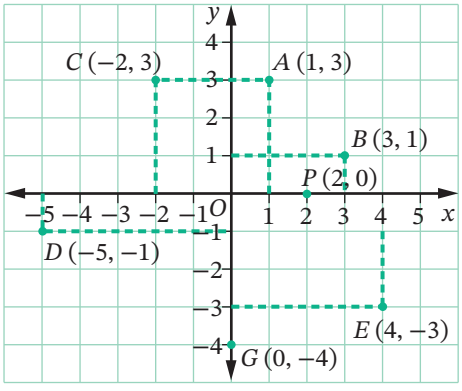
c Find the number of students that could be seated around 18 desks.

d Class 8Y has 29 students. How many desks are needed for them?

The number plane 11.04

A **number plane** is a grid made from a horizontal number line called the ***x*-axis** and a vertical number line called the ***y*-axis**. The number plane is also called a **Cartesian plane**, named after René Descartes (pronounced 'Ren-ay Day-cart'), a French mathematician and philosopher who developed the idea.

Points on the number plane are located using a pair of **coordinates** or an **ordered pair**. Note that order is important with coordinates: $A(1, 3)$ is not the same as $B(3, 1)$.



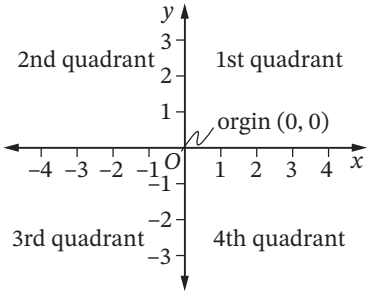
Points on the number plane

In any ordered pair, the first number is called the ***x*-coordinate** and the second number is called the ***y*-coordinate**.

The point (0, 0) is called the **origin**.

To locate a point, always start from (0, 0): the x-coordinate tells you how far to move *across*, the y-coordinate tells you how far to move *up* or *down*.

The number plane is divided evenly into 4 regions called **quadrants**.



Skillsheet
The number plane

Worksheets
The number plane
Plane grid 2
Number plane review
Number plane grid paper
A page of number planes
Plotting points and lines'

Videos
The number plane
Coordinate geometry: Descartes

Foundation Standard Complex

Puzzle
Coordinates code puzzle

Presentation
Graph coordinates

Technology
Plotting points

Spreadsheet
Points plotter

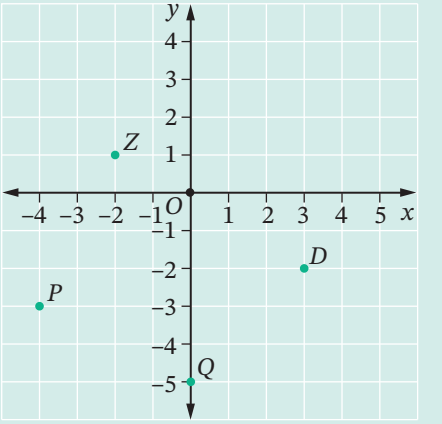
Worksheet
Number plane writing activity

Example 5

a Write the coordinates of point D and state what quadrant it is in.

b Write the point with coordinates $(-4, -3)$ and state what quadrant it is in.

c Write the coordinates of point Q.



Solution

a $D(3, -2)$ is in the 4th quadrant.

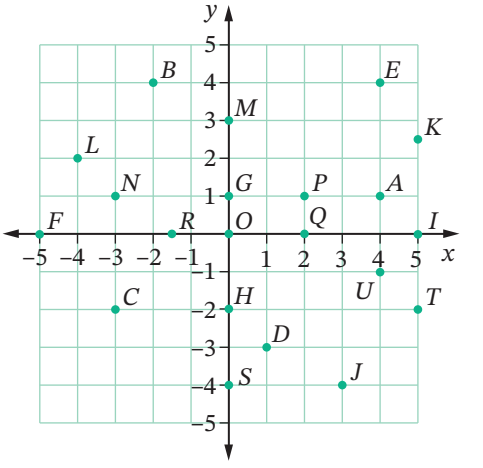
b $(-4, -3)$ is the point P, which is in the 3rd quadrant

c Point Q is 0 on the x-axis and -5 on the y-axis, so its coordinates are $(0, -5)$.

EXERCISE 11.04 ANSWERS ON P. 606

The number plane UFRC

Questions **1** to **5** refer to this diagram.



EXAMPLE 5

1 Name the point with each pair of coordinates.

a $(2, 1)$ **b** $(-3, 1)$

c $(5, 2\frac{1}{2})$ **d** $(5, -2)$

e $(0, 3)$ **f** $(0, -4)$

g $(2, 0)$ **h** $(-1\frac{1}{2}, 0)$

i $(-4, 2)$ **j** $(4, -1)$

2 Write the coordinates of each point. C

a A **b** B **c** C

d D **e** E **f** F

g G **h** H **i** I **j** J

3 In which quadrant does each point lie? C

a N **b** T **c** E **d** L **e** C

4 List all of the points that are on the x-axis and write their coordinates. C

5 List all of the points that are on the y-axis and write their coordinates. C

6 If a point has a positive x-coordinate and a negative y-coordinate, which quadrant is it in? Select the correct answer **A**, **B**, **C** or **D**. R

A 1st quadrant **B** 2nd quadrant **C** 3rd quadrant **D** 4th quadrant

Foundation Standard Complex

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7 Determine the quadrant in which each point lies. R C

a $(3, -5)$ **b** $(-2, -4)$ **c** $(-8, 1)$ **d** $(5, 4)$

8 **a** Draw a number plane, with both axes extending from -8 to 8, or use the link to print Plane grid 2.

b Plot and label these points.

$A(3, 0)$ $B(-2, 2)$ $C(4, 5)$ $D(3\frac{1}{2}, -5)$ $E(-1, 7)$

$F(0, -4)$ $G(-6, 0)$ $H(\frac{1}{2}, -7\frac{1}{2})$ $I(-2, 6)$ $J(7, 0)$

9 **a** Draw a number plane, with both axes extending from -3 to 3, or use the link to print Number plane grid paper.

b Mark these points.

$A(0, 1)$, $B(2, -1)$, $C(-1, 2)$, $D(-2, 3)$, $E(3, -2)$ $F(1, 0)$.

c What do you notice about the points? Check this with your ruler. C

d Points that lie on the same straight line are said to be **collinear**. Write down the coordinates of any 4 points on a number plane that are collinear.

Worksheets
Plane grid 2

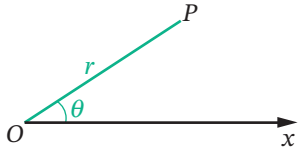
Number plane grid paper

11.04

Did you know?

Polar coordinates

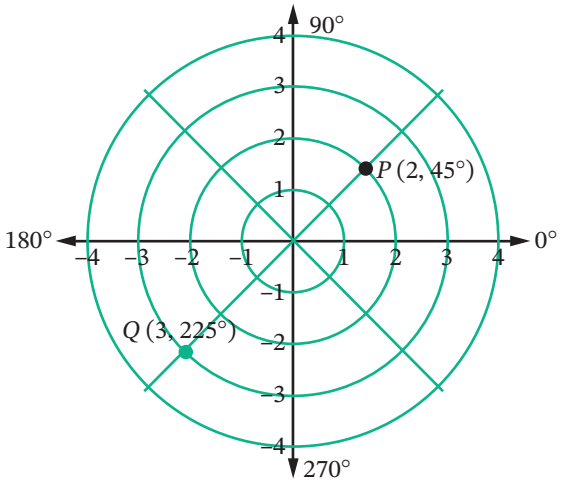
The number plane is not the only way to represent the position of a point. The **polar coordinates** of a point give its position in terms of a distance from the origin and angle measured from the x-axis.



We say that (r, θ) are the polar coordinates of the point P, where r is the distance P is from the origin O and θ is the angle between Ox and OP. A polar grid shows both distance and angle. For example, the grid below shows the coordinates of P as $(2, 45°)$ and Q as $(3, 225°)$.

Some of the real-life uses of polar coordinates include avoiding collisions between vessels and other ships, calculating the flow of ground water and guiding industrial robots.

On polar grid paper, plot the points $L(2, 90°)$, $M(4, 135°)$ and $N(0, 270°)$



polar grid

Foundation Standard Complex

Quiz
Mental skills 11

Mental skills 11: Maths without calculators ANSWERS ON P. 606

The unitary method with percentages

The unitary method is used when you are only given a *percentage* of an amount and you need to find the amount. It is called the unitary method because we find 1% of the amount first, then multiply that by 100 to find the whole (100%).

1 Study each example.

a If 8% of a number is 24, what is the number?

8% of the number = 24

∴ 1% of the number = 24 ÷ 8 = 3

∴ 100% of the number = 3 × 100 = 300.

The number is 300. *Check*: 8% × 300 = 24

b If 15% of an amount is $90, what is the whole amount?

15% of the amount = $90

∴ 1% of the amount = $90 ÷ 15 = $6

∴ 100% of the amount = $6 × 100 = $600.

The amount is $600. *Check*: 15% × $600 = $90

2 Find the whole amount if:

a 5% of the amount is $35

b 11% of the amount is $88

c 20% of the amount is 80

d 6% of the amount is 42

e 90% of the amount is $270

f 15% of the amount is $60

g 40% of the amount is 100

h 120% of the amount is $360

i 25% of the amount is $75

j 8% of the amount is 40

11.05 Graphing number patterns

Video
Graphing number patterns

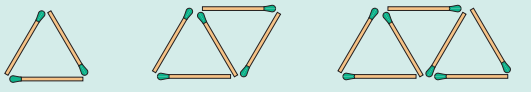
Worksheets
Number plane grid paper

A page of number planes

Graphing tables of values

Example 6

Consider this pattern of matchsticks used to make triangles.



a Copy and complete the table of values for this pattern.

No. of triangles, x	1	2	3	4	5
No. of matchsticks, y					

b Find the formula for this table of values.

c Graph this table of values.

Solution

a

No. of triangles, x	1	2	3	4	5
No. of matchsticks, y	3	5	7	9	11

b Bottom row of y-values increase by 2, so the multiplier in the formula is 2, and the equation is of the form $y = 2x$ ____.

Choose (2, 5) from the table to find the missing number.

When $x = 2$, $y = 2 \times 2 + 1 = 5$.

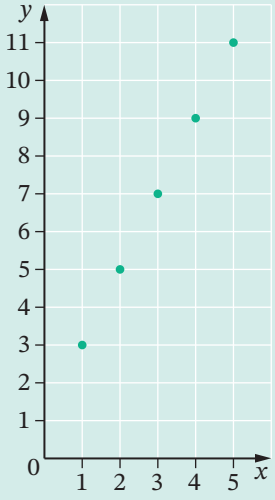
The formula is $y = 2x + 1$.

c Each column in the table forms an ordered pair.

No. of triangle, x	1	2	3	4	5
No. of matchsticks, y	3	5	7	9	11

(1, 3) (2, 5) (3, 7) (4, 9) (5, 11)

Graph each ordered pair on a number plane.



Note that the points are **collinear**, that is, they lie on a straight line. When this happens, we say that the relationship or pattern is **linear**.

- **linear** = involving a line
- **collinear points** = points that lie on a straight line

EXERCISE 11.05 ANSWERS ON P. 606

Graphing number patterns U F R C

1 Find the formula for this table of values. Select the correct answer **A**, **B**, **C** or **D**. R C

x	1	2	3	4	5
y	2	6	10	14	18

A $y = 3x$ **B** $y = 2x + 2$ **C** $y = 4x - 2$ **D** $y = 4x$

2 This table of values shows the height of a tree as it ages. Graph the values and determine whether there is a linear pattern between the age and height of the tree. R C

Age, x (years)	5	10	15	20	25
Height, y (metres)	10	12.5	17.5	21	26

EXAMPLE 6

Foundation Standard Complex

3 For each pattern below: R C

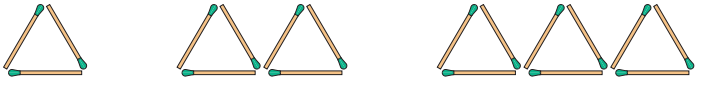
i copy and complete this table

No. of shapes, x	1	2	3	4	5
No. of matchsticks, y					

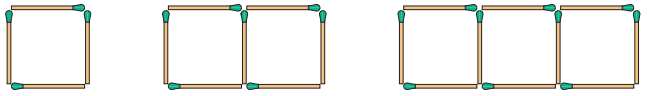
ii find the formula for the table of values

iii graph the table of values.

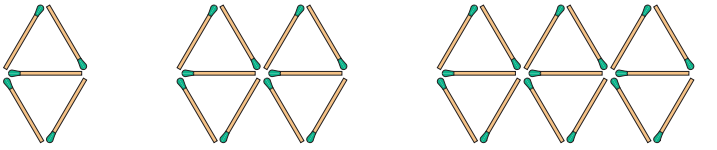
a



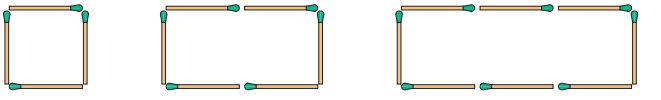
b



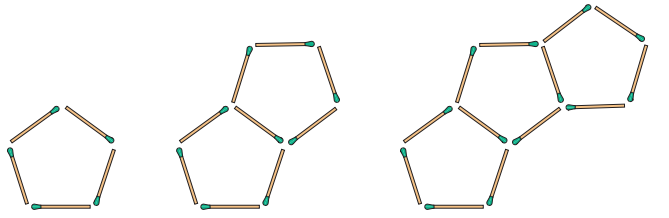
c



d



e



4 Does each graph in question **3** show a linear relationship?

5 Cans in supermarkets are often stacked neatly in triangles. Consider the relationship between the number of cans on the bottom row, x and the total number of cans, y. R C

Shutterstock.com/Giuliano Del Moretto

a Copy and complete this table.

Length of triangle, x	1	2	3	4	5	6	7
No. of cans, y							

b Graph this table on a number plane.

c Is the relationship linear?

d Which formula matches the table of values?
Select the correct answer **A**, **B** or **C**.

A $y = x(x-1)$ **B** $y = \frac{x+1}{2}$ **C** $y = \frac{x(x+1)}{2}$

Foundation Standard Complex

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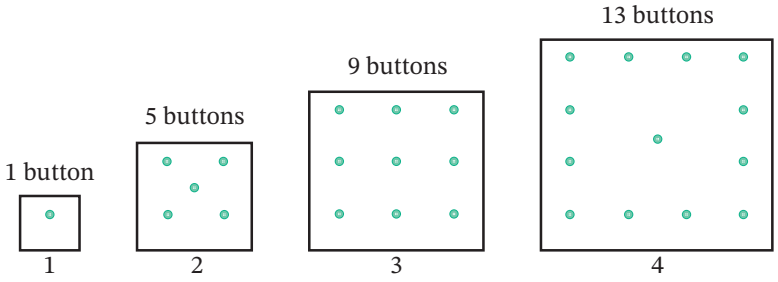
6 The following pattern is made with buttons. R C



a Complete a table of values showing the relationship between the length of the square (S) and the number of buttons (B).

b Find a rule for S in terms of B.

c Use your rule to find the number of buttons required to make a pattern for a square of length:

i 10 **ii** 24

d Kovo has 106 buttons. What is the length of the biggest complete square that she can build?

Graphing linear equations 11.06

A **linear equation** is an equation or formula whose graph on a number plane is a **straight line**.

Graphing linear equations

- complete a table of values
- graph the table of values on a number plane
- rule a line through the points and label the line with the equation.

Example 7

Graph $y = x + 1$ on a number plane.

Solution

Complete a table of values. Choose x-values close to 0 for easy calculation and graphing.

x	−2	−1	0	1	2
y	−1	0	1	2	3

Graph the table of values on a number plane.

Rule a straight line through the points, place arrows at each end, and label the line with its equation.

Worksheets
Number plane grid paper
A page of lines

Puzzles
Matching linear equations
Graphing functions

Interactive
Graphing lines

Videos
Linear functions
Graphing linear equations

Foundation Standard Complex

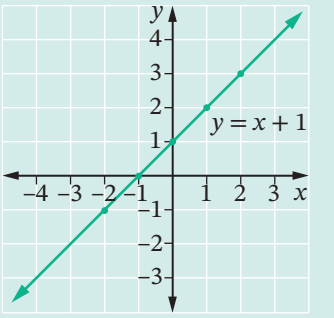
Technology
Graphing straight lines: Finding intercepts

Spreadsheet
Linear functions plotter

Note:

- Every pair of values that follows the linear equation $y = x + 1$ lies on this line, for example, (3, 4), $(-4, -3)$, $(1\frac{1}{2}, 2\frac{1}{2})$, (5, 6), (100, 101), $(-\frac{1}{2}, \frac{1}{2})$.
- Every point on the line follows the linear equation $y = x + 1$.
- There are an infinite number of points that satisfy the rule.

For these reasons, the line is infinite so we must draw arrows at each end.



Video
Graphing linear equations

Example 8

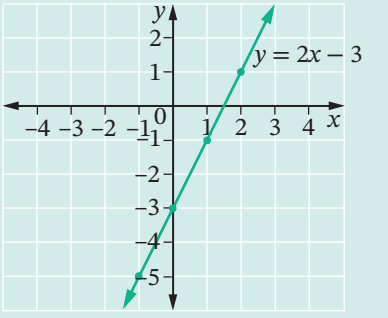
a Graph $y = 2x - 3$ on a number plane.

b Test whether the point (5, 8) lies on the line $y = 2x - 3$.

Solution

a

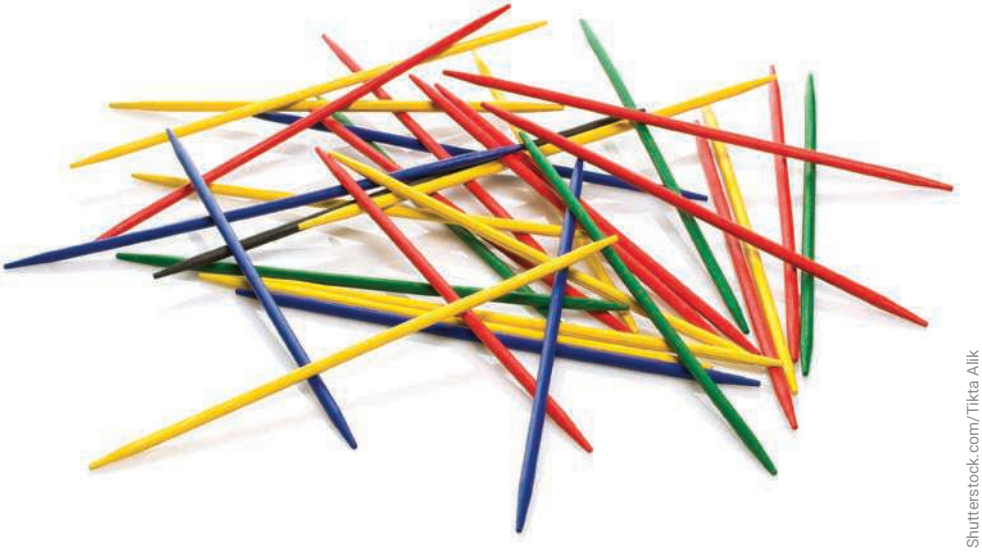
x	−1	0	1	2
y	−5	−3	−1	1



b We could extend the line $y = 2x - 3$ to see if (5, 8) lies on it, but a simpler way is to check whether (5, 8) follows or **satisfies** the formula $y = 2x - 3$. If it does, then when $x = 5$, $y = 8$.

When $x = 5$, $y = 2 \times 5 - 3 = 7 \neq 8$

$\therefore$ (5, 8) does not lie on the line. (In fact, (5, 7) does)



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EXERCISE 11.06 ANSWERS ON P. 607

Graphing linear equations UFRC

1 For each linear equation, copy and complete the table of values and graph the equation on a number plane. C

a $y = x + 3$

x	−1	0	1	2
y				

b $y = x - 2$

x	0	1	2	3
y				

c $y = 2x$

x	−1	0	1	2
y				

d $y = \frac{x}{2}$

x	−1	0	1	2
y				

e $y = 6 - x$

x	0	1	2	3
y				

f $y = x$

x	−1	0	1	2
y				

2 Graph each linear equation on a number plane.

a $y = 2x + 1$ **b** $y = 2x - 1$ **c** $y = 3x$

d $y = 3x - 2$ **e** $y = 2x + 4$ **f** $y = 3x - 5$

3 Look at the graphs you drew in question **2**. Why are they called **increasing graphs**? C

4 Graph each linear equation on a number plane.

a $y = -x$ **b** $y = -x + 5$ **c** $y = 4 - x$

d $y = -x - 3$ **e** $y = 1 - 3x$ **f** $y = -2x$

5 Look at the graphs you drew in question **4**. Why are they called **decreasing graphs**? C

6 By looking at each linear equation, predict whether its graph will be increasing or decreasing. R C

a $y = 5x - 4$ **b** $y = x + 9$ **c** $y = 3 - x$

d $y = 2x$ **e** $y = -x + 5$ **f** $y = -3x - 1$

7 Test whether each point lies on the graph of the given linear equation by: R

i substituting its coordinates into the linear equation

ii examining the graphs of the lines you drew in questions **1** and **2**

a $(4, 7)\ y = x + 3$ **b** $(-1, -1)\ y = x - 2$

c $(5, 1)\ y = 6 - x$ **d** $(-3, -5)\ y = 2x + 1$

e $(4, 10)\ y = 3x - 2$ **f** $(-2, 1)\ y = 2x + 4$

8 On which line does the point $(1, 3)$ lie? Select the correct answer **A**, **B**, **C** or **D**.

A $y = -x + 3$ **B** $y = x + 4$ **C** $y = 2 - 2x$ **D** $y = 2x + 1$

EXAMPLE 7

Worksheets
Number plane grid paper

A page of number planes

EXAMPLE 8

Technology

Graphing linear equations

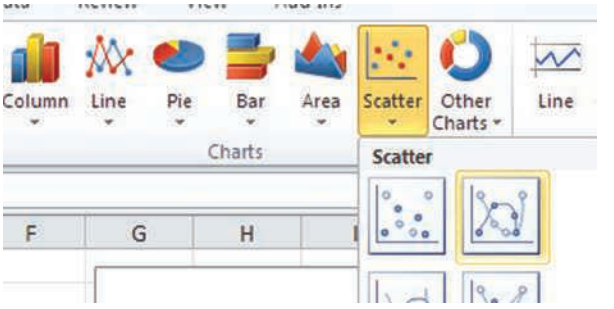
We are going to use a spreadsheet to graph $y = 2x - 3$.

1 Enter these values into a spreadsheet.

	A	B
1	x	y
2	-4	
3	-3	
4	-2	
5	-1	
6	0	
7	1	
8	2	
9	3	
10	4	
11	5	
12		

2 In cell B2, enter the formula **=2*A2−3** to evaluate $2x - 3$ and use **Fill Down** to copy the formula down to cell B11.

3 Highlight columns A and B, then click **Insert** and **Scatter** (with **Smooth Lines and Markers**).



4 Click the bottom-right corner of the graph border to enlarge the graph.

5 Use your graph to predict the value of y when:

a $x = 5$ **b** $x = -6$ **c** $x = -1.5$

6 Use your graph to predict the value of x when:

a $y = 11$ **b** $y = -9$ **c** $y = 14$

Investigation

Special lines

1 Draw a number plane with both axes extending from −4 to 4 and graph each table of values on it.

a

x	−1	0	2	3
y	2	2	2	2

b

x	−2	0	1	3
y	−1	−1	−1	−1

c

x	2	−1	0	−4
y	1	1	1	1

d

x	−2	3	0	4
y	−3	−3	−3	−3

2 **a** What do all of the lines in question **1** have in common?

b What do all of the tables of values in question **1** have in common?

c The formula for the table of values and line in question **1 a** is $y = 2$. It does not have an x in it because the value of y does not depend on x, but is always equal to 2. What are the formulas for the other 3 lines in question **1**?

3 Draw another number plane and graph each table of values on it.

a

x	1	1	1	1
y	−3	−2	0	2

b

x	−2	−2	−2	−2
y	−2	0	1	3

c

x	3	3	3	3
y	−3	2	−2	0

d

x	−3	−3	−3	−3
y	−4	−1	2	4

4 **a** What do all of the lines in question **3** have in common?

b What do all of the tables of values in question **3** have in common?

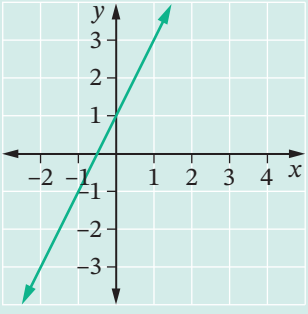
c The formula for the table of values and line in question **3 a** is $x = 1$. What are the formulas for the other 3 lines in question **3**?

Finding the equation of a line

11.07

Example 9

Find the equation of this line.



Solution

First draw a table of values using the coordinates of the points on the line, with consecutive x-values.

x	−2	−1	0	1
y	−3	−1	1	3
		+2	+2	+2

The y-values go up by 2, so the multiplier is 2, and the equation is of the form $y = 2x$ ____.

Choose the point (1, 3) to find the missing number.

When $x = 1$, $y = 2 \times 1 + 1 = 3$.

So the equation is $y = 2x + 1$.

(Check that this is also true for (0, 1)).

Worksheets
Finding linear equations

Puzzles
Matching linear equations

Video
Finding the equation of a line

Video
Horizontal and vertical lines

Horizontal and vertical lines

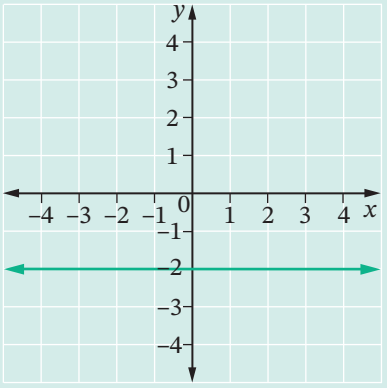
A **horizontal** line is flat and runs across (sideways). The word 'horizontal' comes from 'horizon,' the flat line that is the edge of the Earth as far as your eye can see.

A **vertical** line runs up and down, at right angles to a horizontal line. The word 'vertical' means upright. For example, vertical blinds are made up of columns of slats.

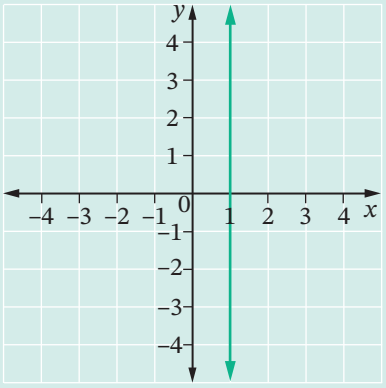
Example 10

Find the equation of each line.

a



b



Solution

a This is a **horizontal** line.

x	−1	0	1	2
y	−2	−2	−2	−2

y is always equal to -2.

So the equation is $y = -2$.

b This is a **vertical** line.

x	1	1	1	1
y	0	1	2	3

x is always equal to 1.

So the equation is $x = 1$.

Horizontal and vertical lines

The **equation of a horizontal line** is of the form $y = c$ (where c is a constant number).

The **equation of a vertical line** is of the form $x = c$ (where c is a constant number).

EXERCISE 11.07 ANSWERS ON P. 609

Finding the equation of a line UFRC

EXAMPLE 9

1 Find the equation that describes this table of values. Select the correct answer **A**, **B**, **C** or **D**.

x	−1	0	1	2
y	3	1	−1	−3

A $y = 2x + 1$ **B** $y = 2x - 1$ **C** $y = x - 2$ **D** $y = -2x + 1$

Foundation Standard Complex

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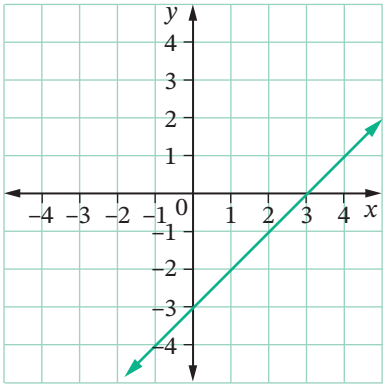
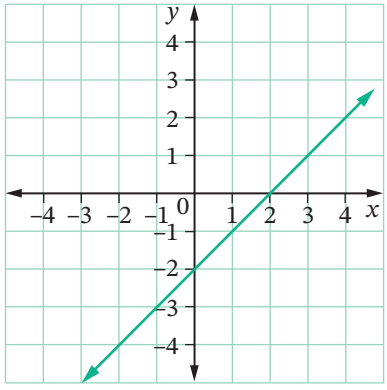
2 For each graph, copy and complete the table of values and use it to find the equation of the line.

a



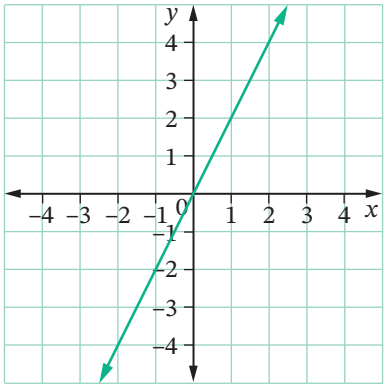
x	−1	0	1	2
y				

b



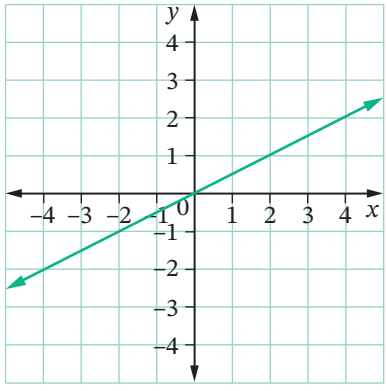
x	−2	−1	0	1
y				

c



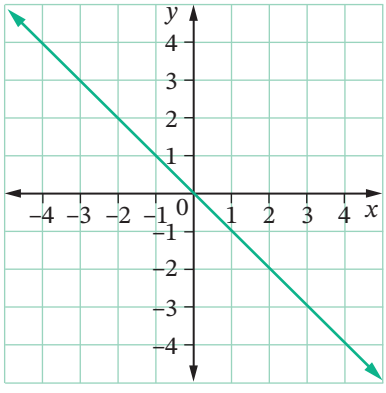
x	−1	0	1	2
y				

d



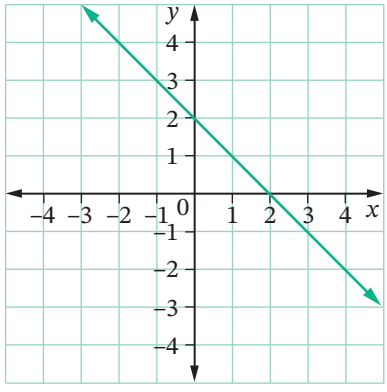
x	0	1	2	3
y				

e



x	−1	0	1	2
y				

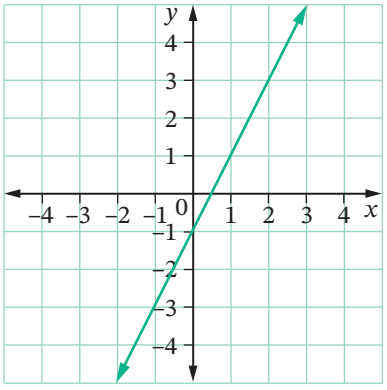
f



x	−3	−2	−1	0
y				

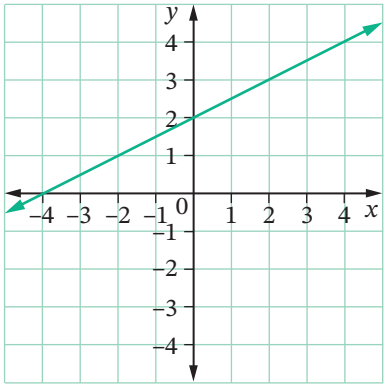
Foundation | Standard | Complex

g



x	−1	0	1	2
y				

h



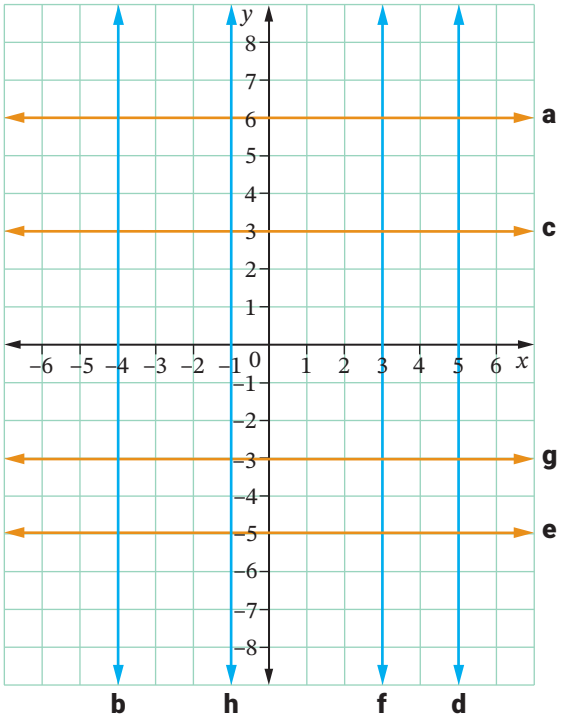
x	−2	−1	0	1
y				

EXAMPLE 10

3 **a** What do the lines represented by $x = 2$, $x = 5$, $x = -6$ and $x = \frac{1}{2}$ have in common? R C

b What do the lines represented by $y = 2$, $y = 5$, $y = -6$ and $y = \frac{1}{2}$ have in common?

4 Write down the equation of each line in the diagram below.



5 Graph each line on the same number plane.

a $x = 4$ **b** $x = -2$ **c** $y = 5$ **d** $y = -1$

e $x = 1$ **f** $y = -3$ **g** $y = 4$ **h** $x = -3$

Foundation Standard Complex

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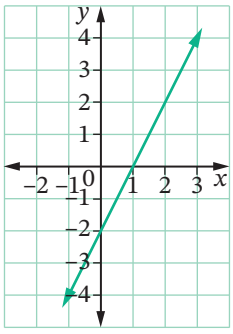
6 What is the equation of this line? Select **A**, **B**, **C** or **D**.



A $y = -x + 2$ **B** $y = 2x - 2$

C $y = 2 - 2x$ **D** $y = x - 2$

7 Harrison read the following points from a line that he drew.

$$(-4, 0), (-3, \tfrac{1}{2}), (-2, 1), (-1, 1\tfrac{1}{2})$$

Find the equation of the line that Harrison drew. R C

11.07

Technology

Comparing linear equations

In this activity, we will use a spreadsheet to compare the graphs of $y = x$, $y = 2x$ and $y = 3x$.

1 Set up this spreadsheet.

	A	B	C	D
1	x	y = x	y = 2x	y = 3x
2	-2			
3	-1			
4	0			
5	1			
6	2			
7				

2 In cell B2 enter **=A2** to calculate $y = x$, then use **Fill Down** to copy this formula down to B6.

3 In cell C2 enter **=2*A2** to calculate $y = 2x$, then **Fill Down** to C6.

4 In cell D2, write a formula to calculate $y = 3x$, then **Fill Down** to D6.

5 Highlight all cells. Click **Insert** and **Scatter (Smooth Lines and Markers)** to graph all 3 lines on the same axes.

6 Compare the 3 lines. What is the same and what is different? Describe any patterns and important features. Write your answers on your spreadsheet or in your workbook.

7 Set up a similar spreadsheet for each set of 3 linear equations shown below.
Use appropriate formulas to complete each column and graph each set of 3 lines.
Compare the 3 lines in each set and describe the pattern and important features of each set.

a

G	H	I	J
x	y = -x	y = -2x	y = -3x
-2			
-1			
0			
1			
2			

b

L	M	N	O
x	y = x + 1	y = x + 2	y = x - 4
-2			
-1			
0			
1			
2			

c

Q	R	S	T
x	y = -x + 1	y = -x + 2	y = -x - 4
-2			
-1			
0			
1			
2			

Foundation Standard Complex

11.08 Comparing linear equations

Puzzles
Matching linear equations

Interactive
Graphing slope-intercept

Video
Linear functions

A **linear equation** such as $y = 2x + 1$ is made up of 2 terms:

- the **variable term**, or the term with the x
- the **constant term**, the number without the x.

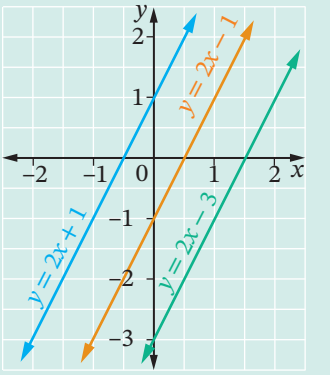
$y = 2x + 1$

The coefficient of x is 2. The constant term is 1.

The number in front of the x is called the **coefficient** of x.

Example 11

The lines $y = 2x + 1$, $y = 2x - 1$ and $y = 2x - 3$ are graphed.



a How are the lines similar?

b How are the linear equations similar?

c The y-intercept of a line is the value at which the line crosses the y-axis. Find the y-intercept of each line in the diagram.

d How are the linear equations different?

Solution

a The lines point in the same direction and have the same slope, they are parallel.

b The linear equations all have the same coefficient of x, namely, 2.

c $y = 2x + 1$: y-intercept $= 1$

$y = 2x - 1$: y-intercept $= -1$

$y = 2x - 3$: y-intercept $= -3$

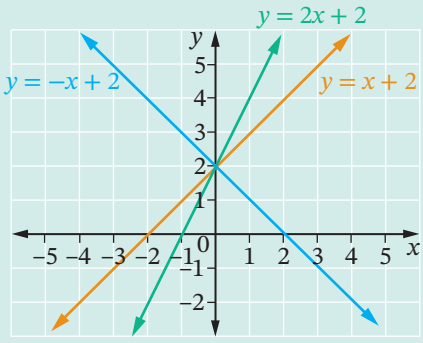
d The linear equations have different constant terms: 1, −1, −3.

iStock.com/Ranimiro Lotufo Neto

The Octavio Frias de Oliveira bridge in São Paulo, Brazil.

Example 12

The lines $y = x + 2$, $y = -x + 2$ and $y = 2x + 2$ are graphed.



a Find the y-intercept of each line.

b How are the linear equations similar?

c How are the lines different?

d How are the linear equations different?

e How does the coefficient of x in a linear equation affect its graph?

f How does the constant term in a linear equation affect its graph?

Solution

a $y = -x + 2$: y-intercept $= 2$

$y = 2x + 2$: y-intercept $= 2$

$y = x + 2$: y-intercept $= 2$

b The linear equations all have the same constant term, 2.

c The lines point in different directions: they are not parallel.

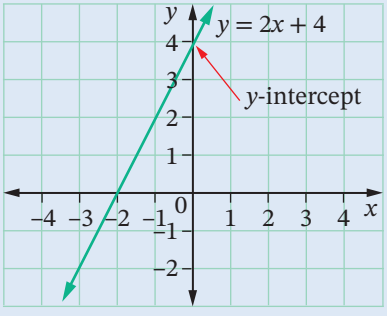
d The linear equations have different coefficients of x: -1, 2 and 1.

e The coefficient of x affects the direction of the graph.

f The constant term gives the y-intercept of the graph.

Features of a linear equation

- In a linear equation, the **coefficient of** x gives the **direction** of the line.
- If 2 or more lines have the same coefficient of x in their equations, then they are parallel.
- The **constant term** gives the y-**intercept** of the line.



EXERCISE 11.08 ANSWERS ON P. 610

Comparing linear equations U F R C

The exercise is best completed using dynamic geometry or graphing software.

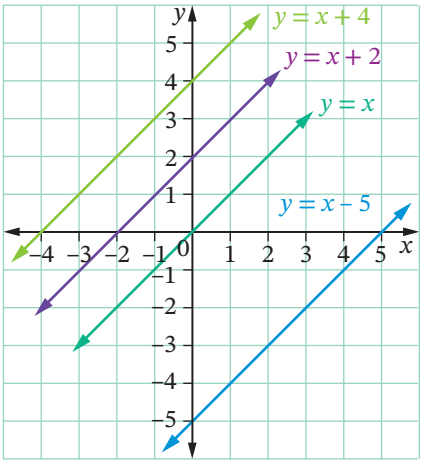
1 For each linear equation, state the coefficient of x and the constant term. C

a $y = 4x + 1$ **b** $y = \frac{1}{2}x - 3$ **c** $y = x - 5$ **d** $y = -x + 6$

EXAMPLE 11

Foundation Standard Complex

2 The lines $y = x + 4$, $y = x + 2$, $y = x$ and $y = x - 5$ are graphed. R C



a How are the lines similar?

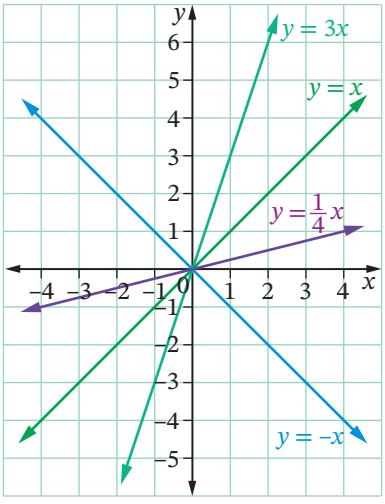
b How are the linear equations similar?

c How are the lines different?

d How are the linear equations different?

EXAMPLE 12

3 The lines $y = \frac{1}{4}x$, $y = x$, $y = 3x$ and $y = -x$ are graphed. R C



a How are the lines similar?

b How are the linear equations similar?

c How are the lines different?

d How are the linear equations different?

e Which line is the steepest? What is the coefficient of x in its equation?

f Which line is the least steep? What is the coefficient of x in its equation?

g How does the size of the coefficient of x in a linear equation affect the steepness of its line?

4 Graph $y = 3x$, $y = 3x - 1$ and $y = 3x + 2$ on the same number plane. R C

a What is the same about all 3 lines?

b What is the same about all 3 linear equations?

c What is different about the lines?

d What is different about the linear equations?

5 Graph $y = x + 1$, $y = 3x + 1$ and $y = \frac{1}{2}x + 1$ on the same number plane. R C

a What is the same about all 3 lines?

b What is the same about all 3 linear equations?

c What is different about the lines?

d What is different about the linear equations?

e What is the coefficient of x in the equation with the steepest line?

6 Graph $y = -x - 2$, $y = -2x - 2$ and $y = -\frac{1}{2}x - 2$ on the same number plane. R C

a How are the lines similar?

b How are the linear equations similar?

c How are the lines different?

d How are the lines different to the lines in question **5**?

e What is the coefficient of x in the equation with the least steepest line?

7 The graphs in question **5** are all increasing, while the graphs in question **6** are all decreasing. What feature of their equations indicates this? R C

8 Which equation below when graphed gives a decreasing line with a y-intercept of 5? Select the correct answer **A**, **B**, **C** or **D**. R C

A $y = 5x$ **B** $y = 5 - x$ **C** $y = 5x - 5$ **D** $y = -5x$

9 Copy and complete each statement. R C

a The equations of parallel lines have the same _____.

b Linear equations with the same constant term have graphs that have the same _____.

c Linear equations with a negative coefficient of x have graphs that are _____.

d The equations of lines with the same y-intercept have the same _____.

e The equations of lines that are decreasing have a _____ coefficient of x.

10 Graph each **non-linear equation** after completing the table of values given. R C

a $y = x^2$

x	−3	−2	−1	0	1	2	3
y							

b $y = \sqrt{x}$

x	0	1	2	4	5	7	9
y							

c $y = x^3$

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y									

d $y = \frac{1}{x}$

x	0.5	1	1.5	2	2.5	3	3.5	4	4.5
y									

11 a Why do you think the equations graphed in question **10** are called non-linear equations? R C

b How are the non-linear equations algebraically different to the linear equations?

11.09 Extension: The gradient of a line

YEAR 9 CORE

The steepness or slope of a line is called the **gradient** of the line, and is calculated as the ratio of the vertical **rise** to the horizontal **run** between any 2 points on the line.

The gradient is represented by the variable m and is measured by counting how many units the line rises (up), for each unit across to the right.

Worksheets
Drawing gradients

A page of lines

Gradient between two points

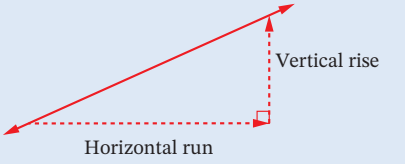
Puzzle
Find the gradient 1

Videos
The gradient of a line

Gradient (slope) of a line

The gradient of a line

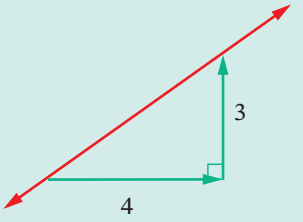
$$m = \frac{\text{vertical rise}}{\text{horizontal run}} = \frac{\text{rise}}{\text{run}}$$



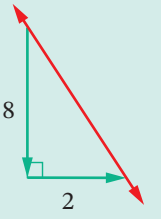
Example 12

Calculate the gradient of each line.

a

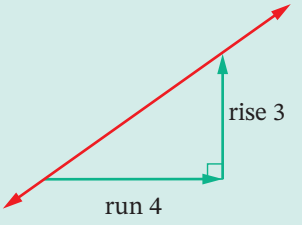


b



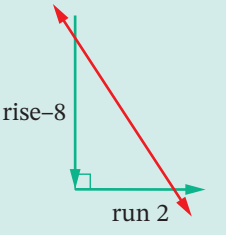
Solution

a $m = \frac{\text{rise}}{\text{run}} = \frac{3}{4}$



Positive gradient,
line leans to the right.

b $m = \frac{\text{rise}}{\text{run}} = \frac{-8}{2} = -4$



Negative gradient,
line leans to the left.
This line has a negative rise or a drop (↓).

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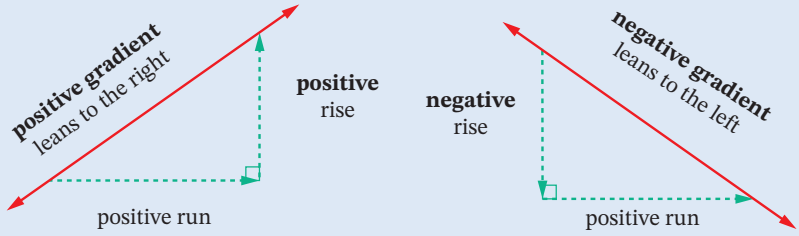
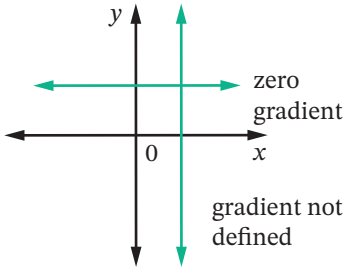
Positive and negative gradients



Note: the run is always **positive**.

A **horizontal** (flat) line has **zero** gradient, as it has no steepness (also no 'rise').

A **vertical** line has infinite steepness and its gradient is **not defined** (also no 'run').

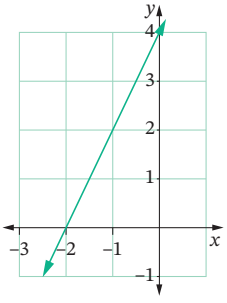


EXERCISE 11.09 ANSWERS ON P. 611

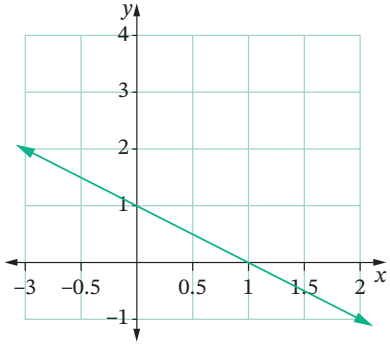
The gradient of a line U F R C

1 State whether the gradient of each line is positive or negative. R C

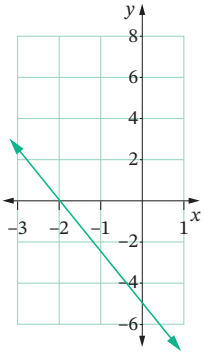
a



b



c



Foundation | Standard | Complex

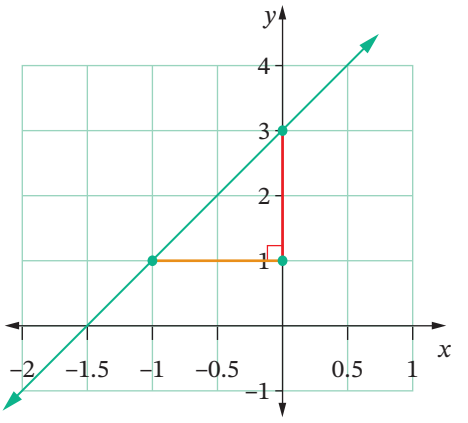
YEAR 9 CORE

EXAMPLE 12

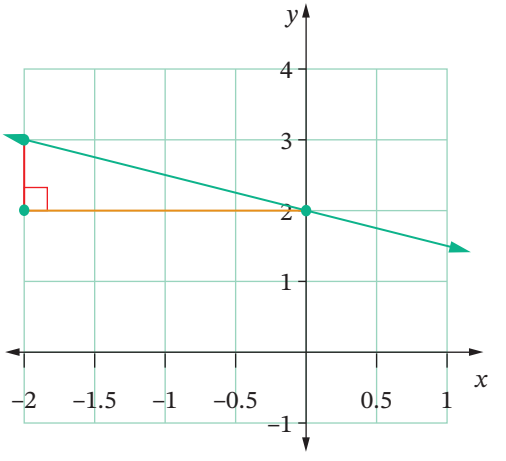
2 Find the gradient of each line in question **1**.

3 Using the given triangles, find the gradient of each line.

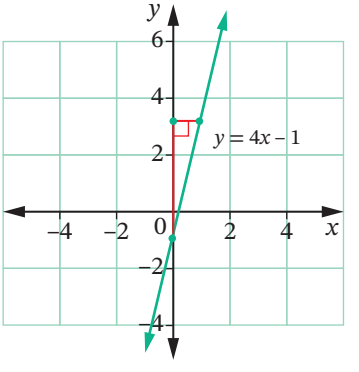
a



b



c



4 Plot each pair of points on a separate number plane and find the gradient of the line that passes through both points.

a (2, 1), (3, 4) **b** (−2, 0), (2, 1) **c** (0, 0), (−4, 0)

d (3, 3), (6, 0) **e** (−3, 1), (−1, 3)

Worksheets
Number plane grid paper

A page of number planes

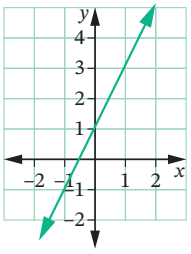
Foundation Standard Complex

9780170479516

Investigation

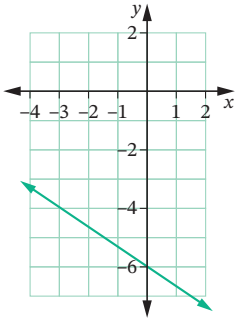
YEAR 9 CORE

Comparing gradients and *y*-intercepts



1 This is the graph of $y = 2x + 1$.

- **a** What is the gradient of the line?
- **b** What is the *y*-intercept?
- **c** How are the above values related to the linear equation $y = 2x + 1$?



2 This is the graph of $y = -\frac{2}{3}x - 6$.

- **a** What is the gradient of the line?
- **b** What is the *y*-intercept?
- **c** Compare your answers with the linear equation $y = -\frac{2}{3}x - 6$.

3 What is the gradient and *y*-intercept of the line with equation $y = 3x - 5$?

11.10

Extension: The linear equation $y = mx + c$

11.10

YEAR 9 CORE

The linear equation $y = mx + c$

The equation of a straight line is $y = mx + c$, where m is the gradient and c is the *y*-intercept.

$y = mx + c$ is called the **gradient-intercept form** of a linear equation.

Example 13

Find the gradient and *y*-intercept of the line with equation:

a $y = 4x + 1$ **b** $y = -\frac{2}{3}x - 5$ **c** $y = 3 - \frac{1}{2}x$ **d** $x + y = 1$

Solution

a $y = 4x + 1$ is in the form $y = mx + c$.

$\therefore$ Gradient $m = 4$ and *y*-intercept $c = 1$.

b For $y = -\frac{2}{3}x - 5$, gradient $m = -\frac{2}{3}$ and *y*-intercept $c = -5$.

c $y = 3 - \frac{1}{2}x$ can be rewritten as $y = -\frac{1}{2}x + 3$.

$\therefore$ Gradient $m = -\frac{1}{2}$ and *y*-intercept $c = 3$.

d $x + y = 1$ can be rewritten as $y = -x + 1$.

$\therefore$ Gradient $m = -1$ and *y*-intercept $c = 1$.

Puzzle
Equation of a line

Interactive
Graphing slope-intercept

Video
Graphing $y = mx + c$

YEAR 9 CORE

Example 14

Graph each equation by finding its gradient and y-intercept.

a $y = 3x - 2$ **b** $y = -\frac{2}{3}x + 1$

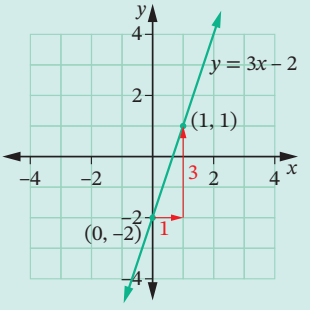
Solution

a $y = 3x - 2$ has a gradient of 3 and a y-intercept of -2

Plot the y-intercept -2 on the y-axis.

Note that $3 = \frac{3}{1} = \frac{up}{right}$. Starting from the y-intercept, move 1 unit right and then 3 units up. This takes you to point $(1, 1)$.

Draw a line through $(1, 1)$ and the y-intercept.

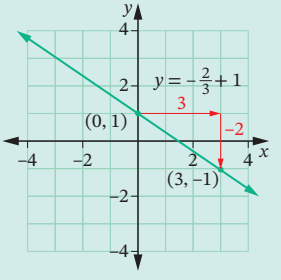


b $y = -\frac{2}{3}x + 1$ has a gradient of $-\frac{2}{3}$ and a y-intercept of 1.

Plot the y-intercept 1 on the y-axis.

Note that $-\frac{2}{3} = \frac{down}{right}$. Starting from the y-intercept, move 3 units right and then 2 units down. This takes you to point $(3, -1)$.

Draw a line through $(3, -1)$ and the y-intercept.



EXERCISE 11.10 ANSWERS ON P. 611

The linear equation $y = mx + c$ UFRC

EXAMPLE 13

1 Write down the gradient and y-intercept of the line with equation:

a $y = 3x + 2$ **b** $y = -2x + 1$ **c** $y = -x + 6$ **d** $y = \frac{1}{3}x - 1$

e $y = -4x - 3$ **f** $y = \frac{2}{5}x + 3$ **g** $y = -\frac{4}{3}x - 8$ **h** $y = \frac{5}{6}x + \frac{1}{2}$

Foundation Standard Complex

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2 Which equation has a gradient of 4 and a y-intercept of 2? Select the correct answer **A**, **B**, **C** or **D**.

A $y = 2x + 4$ **B** $y = 4x - 2$ **C** $y = 2x - 4$ **D** $y = 4x + 2$

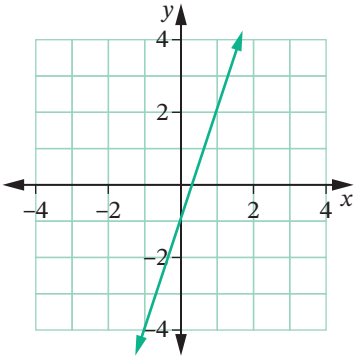
3 Write the equation of the line with:

a a gradient of 3 and y-intercept of 1 **b** a gradient of -2 and y-intercept of 6

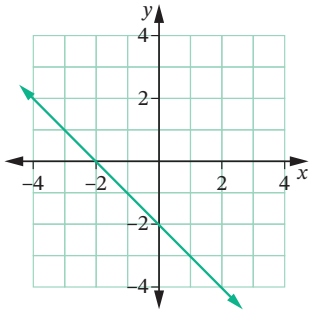
c a gradient of -2 and y-intercept of 11 **d** a gradient of $-\frac{1}{4}$ and y-intercept of -5

4 Find the gradient and y-intercept of each line, then write its equation in the form $y = mx + c$.

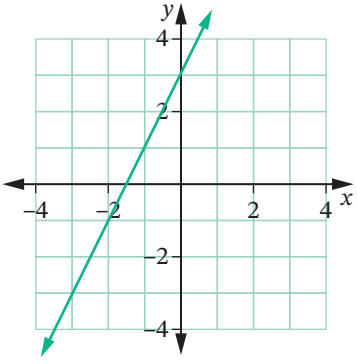
a



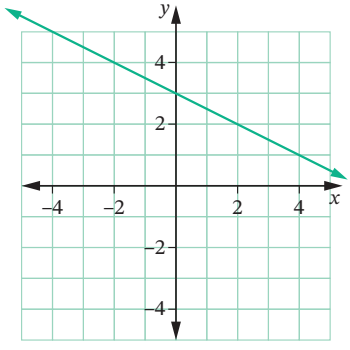
b



c



d



5 Find the gradient and y-intercept of the line with equation:

a $y = 2 - 3x$ **b** $y = -4 - x$ **c** $2x + y = 6$ **d** $3x - y = -1$

6 Graph each equation by finding its gradient and y-intercept.

a $y = 2x + 1$ **b** $y = -\frac{1}{2}x + 3$ **c** $y = -4x$

Worksheets
Number plane grid paper

A page of number planes

EXAMPLE 14

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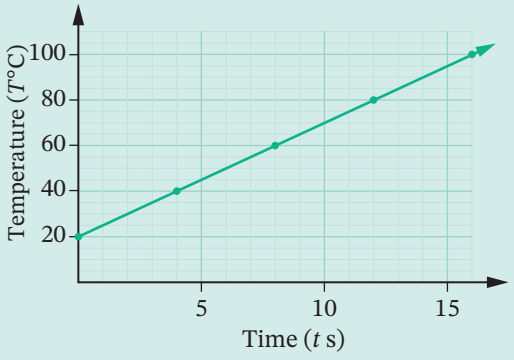
11.11 Applying linear equations

Linear equations can be used to model (represent) real-life situations where a quantity is increasing or decreasing at a constant rate.

Example 15

This table and graph show the temperature, in degrees Celsius, of water being boiled in a kettle. It can be modelled by the linear equation $T = 5t + 20$.

Time (t seconds)	0	4	8	12	16
Temperature (T°C)	20	40	60	80	100



a What is the temperature of the water after 10 seconds?

b Where does the line cross the T-axis? What does this represent?

c When is the temperature of the water 55°C?

d At what rate is the temperature of the water increasing each second?

Solution

a From the graph or equation:

$T = 5 \times 10 + 20$

$= 70$°C

b 20, this represents the starting temperature of the water (20°C), the room temperature.

c From the graph or equation:

$55 = 5t + 20$

$35 = 5t$

$t = 7$ seconds

d From the graph or table, between 0 and 4 seconds, the temperature increased from 20°C to 40°C, an increase of 20°C.

So the temperature of the water is increasing at $\frac{20}{4} = 5$°C per second.

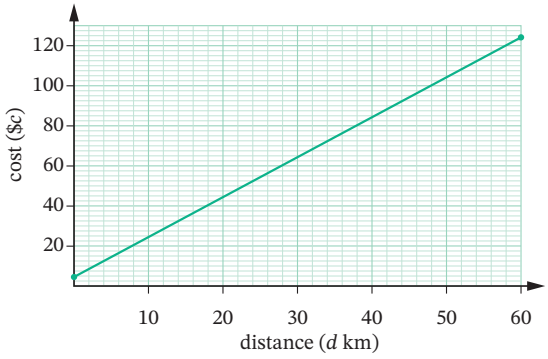
Note that this is also the coefficient of t in the equation $T = 5t + 20$ and the gradient of the line.

EXERCISE 11.11 ANSWERS ON P. 612

Applying linear equations U F PS R C

For this exercise, graphs can be drawn on paper or using dynamic geometry software.

1 The cost of a taxi fare can be represented by the linear equation $c = 2d + 4$ where c is in dollars and d is in kilometres. The graph of the equation is shown below. PS R C



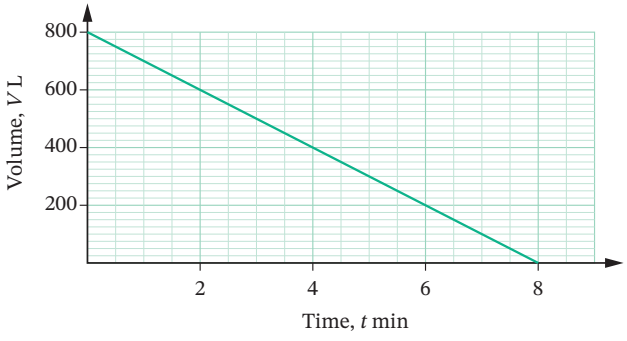
a Find the cost of a trip of 28 km. Use both the equation and the graph.

b If Juan had $100, how far could he travel on the taxi?

c What is the c-intercept of the line? What does this value represent?

d At what rate is the cost increasing per kilometre?

2 This graph shows the volume of water in a 800 L tank being drained over a period of minutes. PS R C



a How long does it take for the tank to empty?

b Find the vertical intercept, c, of the line.

c Find the equation of the line in the form $V = mt + c$, where m and c are numbers.

d How much water was in the tank after 5 minutes?

e How much is the volume of water decreasing in one minute? Is this the same amount for each minute?

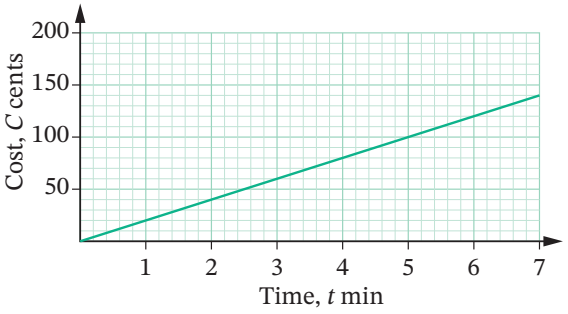
Worksheets
Number plane grid paper
1 cm grid paper

11.11

EXAMPLE 15

Foundation Standard Complex

3 This graph shows the cost of an international phone call, where C is the cost in cents and t is the time in minutes. PS R C



a Find the cost of a phone call which lasts 5 minutes.

b At what rate is the cost increasing each minute?

c Find the vertical intercept of the line.

d Find the equation of the line.

e Alyssa has $1.20, for how long can she talk to her friend?

4 A steel company makes square plates of different sizes. The cost of each plate is calculated using the formula $C = 1.5s + 5$, where C is in dollars and s is in centimetres. PS R C

a Copy and complete this table.

Side length, s cm	10	20	30	40
Cost, $C				

b Graph the line on a number plane.

c In the formula $C = 1.5s + 5$, what does the value of 1.5 represent?

d What does the value of 5 represent?

e Use your graph or equation to determine:

i the cost of manufacturing a plate with side 18 cm

ii the side length of a plate which cost $38

5 Tony kicks a ball towards the goal 60 m away. The distance of the ball from the goal is given by the linear equation $d = 60 - 10t$, where d is in metres and t is in seconds. PS R C

a Copy and complete this table.

Time, t s	0	2	4	6
Distance, d m				

b Graph the linear equation.

c What is the distance of the ball from goal after 5 seconds?

d How long does it take for the ball to reach 30 m from goal?

e How long does it take the ball to reach the goal?

f What is the vertical intercept of the line and what does it represent?

g What is the speed of the ball?

6 Lara finds that the cost of running her car each week is determined by the equation $C = 0.5d + 20$, where C is the cost in dollars and d represents the distance travelled in kilometres. PS R C

a Copy and complete this table.

Distance, d km	0	10	20	30	40
Cost, \$$C$					

b Graph this equation on a number plane.

c What does the vertical intercept of the line represent?

d What does the value of 0.5 in the equation $C = 0.5d + 20$ represent?

e Use your graph or equation to determine:

i the cost of travelling 24 km

ii the distance that has a running cost of \$60

Solving linear equations graphically 11.12

Equations can be solved **algebraically** using the balancing and backtracking methods. However, we can also solve an equation **graphically** by first graphing it on the number plane.

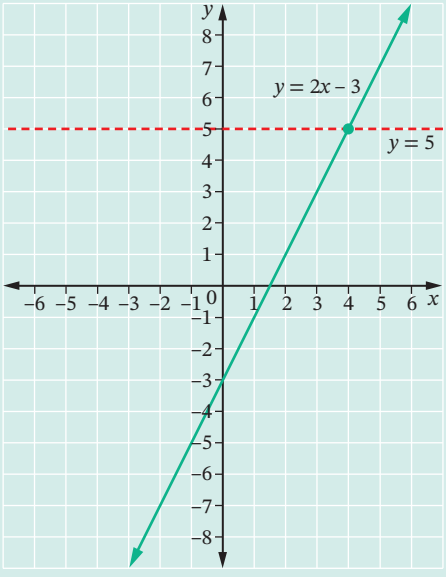
Example 16

Solve the equation $2x - 3 = 5$ graphically.

Solution

Graph $y = 2x - 3$ on a number plane (the LHS of the equation).

x	−1	0	1	2
y	−5	−3	−1	1



For each point in the table and on the line, the y-coordinate is the value of $2x - 3$. So to solve $2x - 3 = 5$, we need to find the point whose y-coordinate is 5.

Draw a dotted horizontal line at $y = 5$ (shown in red on the diagram) and read off the coordinates for the point on $y = 2x - 3$ crossed by this line.

This point is (4, 5), which means that the solution to $2x - 3 = 5$ is $x = 4$.

Check: $2 \times 4 - 3 = 5$

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EXERCISE 11.12 ANSWERS ON P. 612

Solving linear equations graphically U F

This exercise is best completed using dynamic geometry software or a graphing website.

EXAMPLE 16

1 Use the graph in Example **13** above to solve each equation below. Check your solutions.

a $2x - 3 = 3$ **b** $2x - 3 = -7$ **c** $2x - 3 = 0$

2 Graph $y = 2x + 1$ and use it to solve each equation below graphically.

a $2x + 1 = 7$ **b** $2x + 1 = 10$ **c** $2x + 1 = -5$

3 Solve each equation in question **2** algebraically.

4 Graph $y = 2x - 1$ and use it to solve each equation below graphically.

a $2x - 1 = -5$ **b** $2x - 1 = -1$ **c** $2x - 1 = 4$

5 Graph $y = 3x - 2$ and use it to solve each equation graphically.

a $3x - 2 = 7$ **b** $3x - 2 = -11$ **c** $3x - 2 = -2$

6 Graph $y = -x + 1$ and use it to solve each equation graphically.

a $-x + 1 = -3$ **b** $-x + 1 = 4$ **c** $-x + 1 = -1$

7 Graph $y = -2x - 1$ and use it to solve each equation graphically.

a $-2x - 1 = -9$ **b** $-2x - 1 = 0$ **c** $-2x - 1 = 3$

8 Graph $y = \frac{1}{2}x + 3$ and use it to solve each equation graphically.

a $\frac{1}{2}x + 3 = 5$ **b** $\frac{1}{2}x + 3 = 1$ **c** $\frac{1}{2}x + 3 = 2$

Worksheets
Number plane grid paper
A page of number planes

Technology

Intersecting lines

We will graph the 2 straight lines $y = x + 5$ and $y = 3 - x$ using a spreadsheet to find their point of intersection.

	A	B	C
1	x	y = x + 5	y = 3 - x
2	-4		
3	-3		
4	-2		
5	-1		
6	0		
7	1		
8	2		
9	3		
10	4		

1. Set up this spreadsheet.
2. In cell B2, enter **=A2+5**, then **Fill Down** to cell B10.
3. In cell C2, enter **=3−A2**, then **Fill Down** to cell C10.
4. Highlight all 3 columns and click **Insert** and **Scatter (Smooth Lines and Markers)**.
5. From the graph, read the point of intersection of the 2 straight lines.

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6 Find the point of intersection for each pair of straight lines below, by setting up a separate spreadsheet for each pair of lines and using **Insert** and **Scatter (Smooth Lines and Markers)** (as above) to graph each pair of lines on the same set of axes.

a

	A	B	C
1	x	y = 2x - 2	y = -x + 7
2	-4		
3	-3		
4	-2		
5	-1		
6	0		
7	1		
8	2		
9	3		
10	4		

b

	A	B	C
1	x	y = 3x - 6	y = -4x - 6
2	-4		
3	-3		
4	-2		
5	-1		
6	0		
7	1		
8	2		
9	3		
10	4		

c

	A	B	C
1	x	y = 10 - 2x	y = 2x
2	-4		
3	-3		
4	-2		
5	-1		
6	0		
7	1		
8	2		
9	3		
10	4		

Intersecting lines

11.13

Worksheet
Line intersections

Example 17

Graph $y = 2x - 2$ and $y = x + 1$ on the same number plane and use the graphs to solve the equation $2x - 2 = x + 1$.

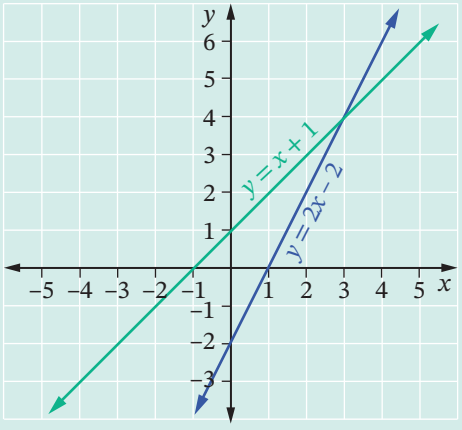
Solution

$y = 2x - 2$

x	−1	0	1	2
y	−4	−2	0	2

$y = x + 1$

x	−1	0	1	2
y	0	1	2	3



To solve $2x - 2 = x + 1$, we need to find the point on both lines where the y-coordinate is the same.

This is the point where the 2 lines intersect.

This point is (3, 4), which means that the solution to $2x - 2 = x + 1$ is $x = 3$.

Check:

LHS $= 2 \times 3 - 2 = 4$

RHS $= 3 + 1 = 4$

LHS = RHS

Worksheets
Number plane grid paper

A page of number planes

EXAMPLE 17

EXERCISE 11.13 ANSWERS ON P. 613

Intersecting lines U F R

This exercise is best completed using dynamic geometry software.

1 By substitution, find the point of intersection of the lines $y = 2x - 1$ and $y = 5 - x$.
Select the correct answer **A**, **B**, **C** or **D**. R

A $(2, -1)$ **B** $(2, 3)$ **C** $(5, -1)$ **D** $(\frac{4}{3}, \frac{5}{3})$

2 Graph each pair of equations and find their point of intersection.

a $y = x$ and $y = -x + 2$ **b** $y = x + 1$ and $y = 2x$

c $y = -x - 1$ and $y = x + 1$ **d** $y = \frac{x}{2} - 2$ and $y = -x + 1$

e $y = 2x + 1$ and $y = x + 3$ **f** $y = -2x - 1$ and $y = x + 5$

3 What is the point of intersection of $y = x + 1$ and $y = -x + 5$? Select **A**, **B**, **C** or **D**.

A $(3, 2)$ **B** $(2, 3)$ **C** $(4, 1)$ **D** $(4, 5)$

4 Solve each equation graphically. R

a $2x - 1 = x + 2$ **b** $3x - 2 = 2x - 1$ **c** $x - 2 = 4 - x$

d $2x = x + 3$ **e** $3x + 2 = x - 1$ **f** $-2x + 3 = 2x + 1$

5 Solve each of the equations above algebraically.

6 These 3 linear equations form a triangle when graphed on the number plane.

$y = x + 2$ $y = \frac{x}{2} - 1$ $y = -3x + 6$

Find the coordinates of the 3 vertices of the triangle. R

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Power plus ANSWERS ON P. 614

1 **a** George receives twice as much as Bill in pocket money. This can be represented by $y = 2x$, where y is George's pocket money and x is Bill's amount of money. Draw the graph of this line on a number plane where the x-axis runs from 0 to 12 and the y-axis from 0 to 24.

b The total of George and Bill's pocket money is \$24. Write an equation that represents this.

c Draw the graph of your equation in part **b** on the same set of axes as part **a**.

d Use your graphs to find George's and Bill's pocket money.

2 Graph each equation accurately and state whether it is linear or non-linear.

a $y = x^3 - 2$ **b** $x + y = 5$ **c** $y = x^2 - 1$

d $y = 4 - x^2$ **e** $x - y = 4$ **f** $y = \frac{4}{x}$

CHAPTER 11 REVIEW

Language of maths

Quiz
Language of maths 11

Puzzle
Linear equations crossword

coefficient	collinear	consecutive	constant term
coordinates	decreasing	formula	graphically
horizontal	increasing	intersection	linear
number plane	origin	quadrant	satisfy
solution	table of values	term	variable
vertical	x-axis	y-axis	y-intercept

1 What is the difference between **linear** and **collinear**?

2 In which quadrant of the number plane are the x- and y-coordinates of a point both negative?

3 What are **consecutive** numbers?

4 What is the 'answer' to an equation called?

5 What does it mean if the coordinates of a point **satisfy** the equation of a line?

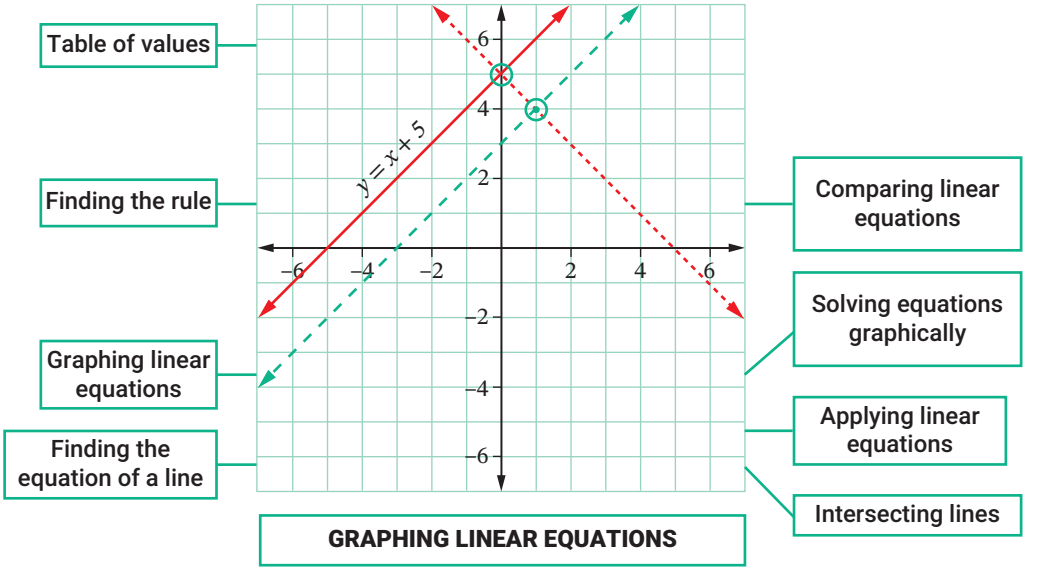
6 For the linear equation $y = 4x + 1$, which is the **constant term**?

Topic summary

Worksheet
Mind map: Graphing linear equations

- Write in your own words what you have learnt about linear equations and their graphs.
- What parts of this topic did you have difficulty with? Discuss them with a friend.
- Where might you use the skills acquired in this chapter?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 11

ANSWERS ON P. 615

1 Copy and complete each table of values. 11.01

a $y = x + 3$

x	−1	0	1	2	3
y					

b $q = 2p - 7$

p	−2	−1	0	1	2
q					

c $d = 5 - c$

c	−1	0	1	2	3
d					

2 Find the formula for each table of values. 11.02

a

x	−1	0	1	2	3
y	−3	−2	−1	0	1

b

m	1	2	3	4	5
p	2	5	8	11	14

c

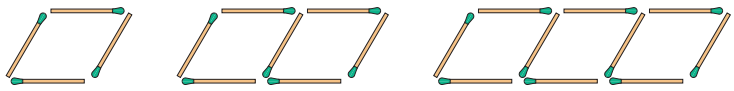
m	0	1	2	3	4
n	10	8	6	4	2

3 For this number pattern: 6, 10, 14, 18, 22, ... 11.03

a find the 7th term

b find a formula for the nth term, T.

4 This pattern of rhombuses is made up of matchsticks. 11.03



a Copy and complete the table.

No. of rhombuses, x	1	2	3	4	5	8	10
No. of matchsticks, y	4						

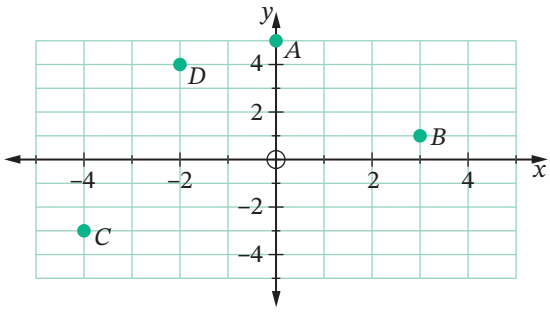
b Write the rule for the pattern in words.

c Write the rule as a formula.

d How many matchsticks are needed to build a pattern of:

i 12 rhombuses? **ii** 15 rhombuses?

5 State the coordinates of the points A, B, C and D. 11.04



6 Plot each point on a number plane. 11.04

a $P(1, -4)$ **b** $Q(-3, -1)$ **c** $R(-1, 2)$

d $S(0, 4)$ **e** $T(2\frac{1}{2}, -2)$ **f** $U(-2, 0)$

Foundation | Standard | Complex

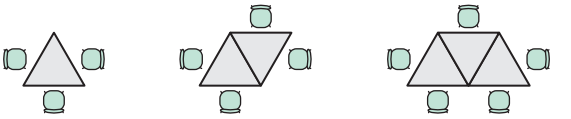
11.04 **7** Which point from question **6**:

a is on the y-axis? **b** is in the 2nd quadrant?

c is on the x-axis? **d** is in the 4th quadrant?

11.05 **8** **a** Copy and complete this table for the pattern of tables and chairs.

Number of tables (x)	1	2	3	4	5
Number of chairs (y)					



b Find the formula for y.

c Graph the table of values.

11.06 **9** Graph each linear equation on a number plane after completing a table of values.

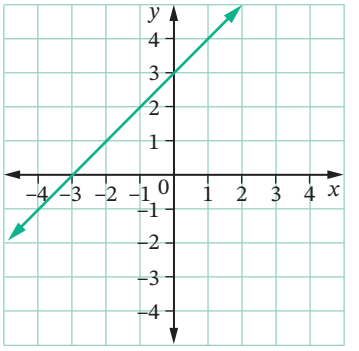
a $y = 2x - 1$ **b** $y = -x + 3$ **c** $y = \frac{1}{2}x - 2$

11.06 **10** Test whether each point lies on the line $y = 3x + 2$.

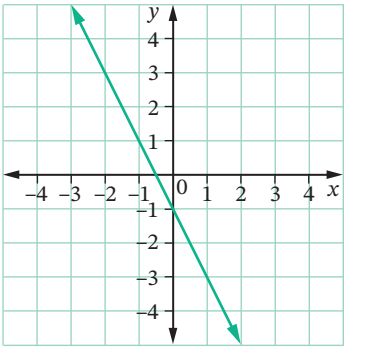
a $(1, 2)$ **b** $(-1, 4)$ **c** $(-2, -4)$ **d** $(0, 2)$

11.07 **11** Find the equation of each line.

a



b



11.07 **12** Graph each linear equation on the same number plane.

a $y = -3$ **b** $x = 4$

11.08 **13** **a** Graph $y = -x + 2$, $y = x + 2$ and $y = 2x + 2$ on the same number plane.

b What is the same about the 3 lines?

c What is the same about the 3 linear equations?

d What is different about the 3 lines?

14 Find the gradient of each line shown in Question **11**.

15 Draw a line through the 2 points (0, –3) and (2, 1) on a number plane and find its gradient.

16 Write the equation of the line with:

a gradient –4 and y-intercept 8 **b** gradient $\frac{-3}{2}$ and y-intercept –6

17 Graph $y = -2x - 1$ on a number plane by finding its gradient and y-intercept.

18 Serge provides waiters for a catering company and charges according to the formula $C = 100 + 30w$, where C is the cost in dollars and w is the number of waiters.

a Copy and complete the table below.

No. of waiters, w	1	2	4	6
Cost, \$$C$				

b Graph this line on the number plane.

c Find the cost of hiring 5 waiters.

d If Annalise paid \$310, how many waiters did she hire?

e What is the cost of each additional waiter?

19 **a** Graph $y = 3x - 4$ and use it to solve $3x - 4 = 5$ graphically.

b Graph $y = 2x + 1$ and use it to solve $2x + 1 = -2$ graphically.

20 Graph $y = 2x$ and $y = x - 2$ and find their point of intersection.

21 Solve the equation $3x - 3 = x + 3$:

a graphically **b** algebraically